MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

HIGH PERFORMANCE EMULATION CONTROL II MODULE, KEY CODE FWA

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic high performance emulation control II module.

1.2 Classification. This specification covers two classes of modules. Classes I and II as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

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3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 1.587 failures per million operating hours for class I or 3.740 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, the failure rate for class I shall be used for reliability prediction.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in class I environments, the failure rate for class I shall be used for reliability prediction.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be FWA.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 112 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

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TABLE I. Module electrical requirements.

Test cond	Parameter 3/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
A1	Isolation	II	S	10	INF	10	INF	10	INF	MOhm
A2	Isolation	II	X	10	INF	10	INF	10	INF	MOhm
B1-B6	Continuity	III	X	0	1	0	1	0	1	Ohm
C	Vcc stress	IV	S	---Module shall pass all remaining tests---						
D1	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D2	Ibd	V	S	0.0	2.1	0.0	2.1	0.0	2.1	mA
D3	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D4	Ibd	V	S	-0.02	0.12	-0.02	0.12	-0.02	0.12	mA
D5	Ibd	V	S	-0.02	1.02	-0.02	1.02	-0.02	1.02	mA
D6,D7	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D8	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D9	Ibd	V	S	-0.02	1.02	-0.02	1.02	-0.02	1.02	mA
D10	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D11	Ibd	V	S	0.0	2.2	0.0	2.2	0.0	2.2	mA
D12	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D13	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D14	Ibd	V	S	0.0	1.1	0.0	1.1	0.0	1.1	mA
D15	Ibd	V	S	0.0	2.2	0.0	2.2	0.0	2.2	mA
D16	Ibd	V	S	0.0	1.1	0.0	1.1	0.0	1.1	mA
D17-D23	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D24	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D25	Ibd	V	S	-0.02	0.12	-0.02	0.12	-0.02	0.12	mA
D26	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D27-D32	Ibd	V	S	-0.02	0.12	-0.02	0.12	-0.02	0.12	mA
D33	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D34	Ibd	V	S	-0.02	0.12	-0.02	0.12	-0.02	0.12	mA
D35,D36	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D37-D39	Ibd	V	S	0.0	1.1	0.0	1.1	0.0	1.1	mA
D40	Ibd	V	S	-0.02	1.12	-0.02	1.12	-0.02	1.12	mA
D41,D42	Ibd	V	S	0.0	0.3	0.0	0.3	0.0	0.3	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 3/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
D43	Ibd	V	5	-0.02	1.02	-0.02	1.02	-0.02	1.02	mA
D44,D45	Ibd	V	5	0.0	1.1	0.0	1.1	0.0	1.1	mA
D46	Ibd	V	5	0.0	1.0	0.0	1.0	0.0	1.0	mA
D47	Ibd	V	5	0.0	2.0	0.0	2.0	0.0	2.0	mA
D48	Ibd	V	5	0.0	0.1	0.0	0.1	0.0	0.1	mA
D49	Ibd	V	5	0.0	1.1	0.0	1.1	0.0	1.1	mA
D50	Ibd	V	5	0.0	1.3	0.0	1.3	0.0	1.3	mA
D51	Ibd	V	5	0.0	1.1	0.0	1.1	0.0	1.1	mA
D52	Ibd	V	5	0.0	0.4	0.0	0.4	0.0	0.4	mA
D53-D55	Ibd	V	5	0.0	0.1	0.0	0.1	0.0	0.1	mA
D56	Ibd	V	5	0.0	0.2	0.0	0.2	0.0	0.2	mA
D57-D62	Ibd	V	5	-0.02	0.12	-0.02	0.12	-0.02	0.12	mA
E1	Ioe	VI	5	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E2	Ios	VI	5	-100.0	-40.0	-100.0	-40.0	-100.0	-40.0	mA
E3	Ioe	VI	5	-100.0	-15.0	-100.0	-15.0	-100.0	-15.0	mA
E4,E5	Ios	VI	5	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E6	Ios	VI	5	-100.0	-15.0	-100.0	-15.0	-100.0	-15.0	mA
E7	Ioe	VI	5	-130.0	-30.0	-130.0	-30.0	-130.0	-30.0	mA
E8	Ios	VI	5	-100.0	-15.0	-100.0	-15.0	-100.0	-15.0	mA
E9-E12	Ioe	VI	5	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E13	Ioe	VI	5	-130.0	-30.0	-130.0	-30.0	-130.0	-30.0	mA
E14	Ios	VI	5	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E15	Ioe	VI	5	-100.0	-40.0	-100.0	-40.0	-100.0	-40.0	mA
E16	Ios	VI	5	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E17,E18	Ios	VI	5	-130.0	-30.0	-130.0	-30.0	-130.0	-30.0	mA
E19,E20	Ioe	VI	5	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E21	Ios	VI	5	-85.0	-15.0	-85.0	-15.0	-85.0	-15.0	mA
E22,E23	Ioe	VI	5	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E24	Ioe	VI	5	-260.0	-30.0	-260.0	-30.0	-260.0	-30.0	mA
E25,E26	Ios	VI	5	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E27	Ioe	VI	5	-100.0	-40.0	-100.0	-40.0	-100.0	-40.0	mA
E28	Ios	VI	5	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 3°C		Class II 4/ -55 ± 5°C 85 ± 3°C		Units
				Min	Max	Min	Max	Min	Max	
F1	Ith	VII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
F2	Ith	VII	X	-1.1	0.07	-1.1	0.07	-1.1	0.07	mA
F3	Ith	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F4	Ith	VII	X	-1.88	-0.54	-1.86	-0.54	-1.88	-0.54	mA
F5	Ith	VII	X	-3.06	-0.51	-3.06	-0.51	-3.06	-0.51	mA
F6	Ith	VII	X	-0.8	0.04	-0.8	0.04	-0.8	0.04	mA
F7	Ith	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F8	Ith	VII	X	-1.0	0.04	-1.0	0.04	-1.0	0.04	mA
F9	Ith	VII	X	-1.31	-0.54	-1.31	-0.54	-1.31	-0.54	mA
F10	Ith	VII	X	-0.25	0.03	-0.25	0.03	-0.25	0.03	mA
F11	Ith	VII	X	-1.7	0.08	-1.7	0.09	-1.7	0.08	mA
F12	Ith	VII	X	-0.25	0.03	-0.25	0.03	-0.25	0.03	mA
F13	Ith	VII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
F14	Ith	VII	X	-2.6	0.07	-2.6	0.07	-2.6	0.07	mA
F15	Ith	VII	X	-5.0	0.14	-5.0	0.14	-5.0	0.14	mA
F16	Ith	VII	X	-0.85	0.03	-0.85	0.03	-0.85	0.03	mA
F17-F19	Ith	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F20	Ith	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F21,F22	Ith	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F23	Ith	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F24	Ith	VII	X	-0.8	0.04	-0.8	0.04	-0.8	0.04	mA
F25	Ith	VII	X	-1.88	-0.54	-1.86	-0.54	-1.88	-0.54	mA
F26	Ith	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F27,F28	Ith	VII	X	-1.88	-0.54	-1.86	-0.54	-1.86	-0.54	mA
F29-F32	Ith	VII	X	-1.26	-0.54	-1.26	-0.54	-1.26	-0.54	mA
F33	Ith	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F34	Ith	VII	X	-1.28	-0.54	-1.28	-0.54	-1.28	-0.54	mA
F35,F36	Ith	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F37-F39	Ith	VII	X	-0.85	0.03	-0.85	0.03	-0.85	0.03	mA
F40	Ith	VII	X	-3.26	-0.49	-3.26	-0.49	-3.26	-0.49	mA
F41	Ith	VII	X	-1.68	0.08	-1.68	0.08	-1.68	0.08	mA
F42	Ith	VII	X	-1.8	0.08	-1.8	0.08	-1.8	0.08	mA

- See footnotes at end of table.

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Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 55 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
F43	Ith	VII	X	-1.31	-0.54	-1.31	-0.54	-1.31	-0.54	mA
F44,F45	Ith	VII	X	-0.85	0.05	-0.85	0.05	-0.85	0.05	mA
F46	Ith	VII	X	-0.25	0.03	-0.25	0.03	-0.25	0.03	mA
F47	Ith	VII	X	-0.5	0.05	-0.5	0.05	-0.5	0.05	mA
F48	Ith	VII	X	-0.8	0.02	-0.8	0.02	-0.8	0.02	mA
F49	Ith	VII	X	-2.6	0.07	-2.6	0.07	-2.6	0.07	mA
F50	Ith	VII	X	-2.05	0.09	-2.05	0.09	-2.05	0.09	mA
F51	Ith	VII	X	-0.05	0.05	-0.05	0.05	-0.05	0.05	mA
F52	Ith	VII	X	-2.28	0.1	-2.28	0.1	-2.28	0.1	mA
F53,F54	Ith	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F55	Ith	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F56	Ith	VII	X	-0.8	0.04	-0.8	0.04	-0.8	0.04	mA
F57-F59	Ith	VII	X	-1.66	-0.54	-1.66	-0.54	-1.66	-0.54	mA
F60-F62	Ith	VII	X	-1.26	-0.54	-1.26	-0.54	-1.26	-0.54	mA
G1	Iil	VIII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
G2	Iil	VIII	X	-1.1	0.07	-1.1	0.07	-1.1	0.07	mA
G3	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G4	Iil	VIII	X	-1.66	-0.54	-1.66	-0.54	-1.66	-0.54	mA
G5	Iil	VIII	X	-3.06	-0.51	-3.06	-0.51	-3.06	-0.51	mA
G6	Iil	VIII	X	-0.8	0.04	-0.8	0.04	-0.8	0.04	mA
G7	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G8	Iil	VIII	X	-1.0	0.04	-1.0	0.04	-1.0	0.04	mA
G9	Iil	VIII	X	-1.31	-0.54	-1.31	-0.54	-1.31	-0.54	mA
G10	Iil	VIII	X	-0.25	0.03	-0.25	0.03	-0.25	0.03	mA
G11	Iil	VIII	X	-1.7	0.09	-1.7	0.09	-1.7	0.09	mA
G12	Iil	VIII	X	-0.25	0.03	-0.25	0.03	-0.25	0.03	mA
G13	Iil	VIII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
G14	Iil	VIII	X	-2.6	0.07	-2.6	0.07	-2.6	0.07	mA
G15	Iil	VIII	X	-5.0	0.14	-5.0	0.14	-5.0	0.14	mA
G16	Iil	VIII	X	-0.05	0.05	-0.05	0.05	-0.05	0.05	mA
G17-G19	Iil	VIII	X	-0.5	0.02	-0.5	0.02	-0.5	0.02	mA
G20	Iil	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G21,G22	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G23	Iil	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA

- See footnotes at end of table.

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Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 + 5°C 60 ± 5°C		Class II 4/ -55 + 5°C 85 ∓ 5°C		Units
				Min	Max	Min	Max	Min	Max	
G24	Iil	VIII	X	-0.8	0.04	-0.8	0.04	-0.8	0.04	mA
G25	Iil	VIII	X	-1.66	-0.54	-1.66	-0.54	-1.66	-0.54	mA
G26	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G27,G28	Iil	VIII	X	-1.66	-0.54	-1.66	-0.54	-1.66	-0.54	mA
G29-G32	Iil	VIII	X	-1.26	-0.54	-1.26	-0.54	-1.26	-0.54	mA
G33	Iil	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G34	Iil	VIII	X	-1.26	-0.54	-1.26	-0.54	-1.26	-0.54	mA
G35,G36	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G37-G39	Iil	VIII	X	-0.05	0.05	-0.05	0.05	-0.05	0.05	mA
G40	Iil	VIII	X	-3.26	-0.49	-3.26	-0.49	-3.26	-0.49	mA
G41	Iil	VIII	X	-1.60	0.00	-1.60	0.00	-1.60	0.00	mA
G42	Iil	VIII	X	-1.8	0.06	-1.8	0.06	-1.8	0.06	mA
G43	Iil	VIII	X	-1.31	-0.54	-1.31	-0.54	-1.31	-0.54	mA
G44,G45	Iil	VIII	X	-0.05	0.05	-0.05	0.05	-0.05	0.05	mA
G46	Iil	VIII	X	-0.25	0.03	-0.25	0.03	-0.25	0.03	mA
G47	Iil	VIII	X	-0.5	0.05	-0.5	0.05	-0.5	0.05	mA
G48	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G49	Iil	VIII	X	-2.6	0.07	-2.6	0.07	-2.6	0.07	mA
G50	Iil	VIII	X	-2.05	0.08	-2.05	0.08	-2.05	0.08	mA
G51	Iil	VIII	X	-0.05	0.05	-0.05	0.05	-0.05	0.05	mA
G52	Iil	VIII	X	-2.28	0.1	-2.28	0.1	-2.28	0.1	mA
G53,G54	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G55	Iil	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G56	Iil	VIII	X	-0.8	0.04	-0.8	0.04	-0.8	0.04	mA
G57-G59	Iil	VIII	X	-1.66	-0.54	-1.66	-0.54	-1.66	-0.54	mA
G60-G62	Iil	VIII	X	-1.26	-0.54	-1.26	-0.54	-1.26	-0.54	mA
H1-H28	Voh	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J28	Vol	X	X	0.0	0.5	0.0	0.5	0.0	0.5	V
K1-K5	Iozh	XI	X	-0.05	0.05	-0.05	0.05	-0.05	0.05	mA
K8-K10	Iozl	XI	X	-0.05	0.05	-0.05	0.05	-0.05	0.05	mA
L1-L3	Vic	XII	9	-1.2	0.0	-1.2	0.0	-1.2	0.0	V

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 3/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
L4	Vlc	XII	9	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L5,L6	Vlc	XII	6	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L7-L10	Vlc	XII	9	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L11-L16	Vlc	XII	6	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L17	Vlc	XII	9	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L18,L19	Vlc	XII	9	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L20	Vlc	XII	6	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L21,L22	Vlc	XII	9	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L23	Vlc	XII	9	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L24,L25	Vlc	XII	6	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L26-L28	Vlc	XII	9	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L29,L30	Vlc	XII	6	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L31-L34	Vlc	XII	9	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L35-L38	Vlc	XII	6	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L39	Vlc	XII	9	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L40,L41	Vlc	XII	6	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L42,L43	Vlc	XII	9	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
M1	PSFC	XIII	6	5.1	9.5	5.1	9.5	5.1	9.5	uF
M2	PSFC	XIII	9	7.8	14.6	7.8	14.6	7.8	14.6	uF
N	Icc	XIV	X	856	1284	856	1284	856	1284	mA
P1,P2	Functional	XV	X	See table XV						
Q1,Q2	Dynamic	XVI	9	See table XVI						
Q3,Q4	Dynamic	XVI	0	See table XVI						
R1	Cin	XVIII	0	4	100	4	100	4	100	pF
R2	Cin	XVIII	0	6	125	6	125	6	125	pF
R3	Cin	XVIII	0	2	50	2	50	2	50	pF
R4,R5	Cin	XVIII	0	4	125	4	125	4	125	pF
R6,R7	Cin	XVIII	0	2	75	2	75	2	75	pF
R8	Cin	XVIII	0	4	75	4	75	4	75	pF

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 8 ± 5°C 80 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
R9	Cin	XVIII	0	4	100	4	100	4	100	pF
R10	Cin	XVIII	0	2	75	2	75	2	75	pF
R11	Cin	XVIII	0	8	125	8	125	8	125	pF
R12,R13	Cin	XVIII	0	2	75	2	75	2	75	pF
R14	Cin	XVIII	0	4	75	4	75	4	75	pF
R15	Cin	XVIII	0	8	125	8	125	8	125	pF
R16	Cin	XVIII	0	4	100	4	100	4	100	pF
R17	Cin	XVIII	0	2	50	2	50	2	50	pF
R18,R19	Cin	XVIII	0	2	75	2	75	2	75	pF
R20	Cin	XVIII	0	2	50	2	50	2	50	pF
R21-R23	Cin	XVIII	0	2	75	2	75	2	75	pF
R24,R25	Cin	XVIII	0	4	75	4	75	4	75	pF
R26	Cin	XVIII	0	2	75	2	75	2	75	pF
R27,R28	Cin	XVIII	0	4	50	4	50	4	50	pF
R29	Cin	XVIII	0	4	75	4	75	4	75	pF
R30	Cin	XVIII	0	4	50	4	50	4	50	pF
R31	Cin	XVIII	0	4	75	4	75	4	75	pF
R32	Cin	XVIII	0	4	50	4	50	4	50	pF
R33	Cin	XVIII	0	2	50	2	50	2	50	pF
R34	Cin	XVIII	0	4	75	4	75	4	75	pF
R35,R36	Cin	XVIII	0	2	100	2	100	2	100	pF
R37-R38	Cin	XVIII	0	4	100	4	100	4	100	pF
R40	Cin	XVIII	0	6	175	6	175	6	175	pF
R41,R42	Cin	XVIII	0	6	125	6	125	6	125	pF
R43,R44	Cin	XVIII	0	4	125	4	125	4	125	pF
R45	Cin	XVIII	0	2	100	2	100	2	100	pF
R46	Cin	XVIII	0	2	75	2	75	2	75	pF
R47	Cin	XVIII	0	4	75	4	75	4	75	pF
R48	Cin	XVIII	0	2	75	2	75	2	75	pF
R49	Cin	XVIII	0	4	75	4	75	4	75	pF
R50	Cin	XVIII	0	8	100	8	100	8	100	pF
R51	Cin	XVIII	0	4	75	4	75	4	75	pF
R52	Cin	XVIII	0	6	125	6	125	6	125	pF
R53,R54	Cin	XVIII	0	2	50	2	50	2	50	pF

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
R55	Cin	XVIII	0	2	75	2	75	2	75	pF
R56	Cin	XVIII	0	2	100	2	100	2	100	pF
R57,R58	Cin	XVIII	0	4	75	4	75	4	75	pF
R59	Cin	XVIII	0	4	50	4	50	4	50	pF
R60-R62	Cin	XVIII	0	4	75	4	75	4	75	pF
S1-S3	Input clamp diode latch-up	XIX	0	-1.2	2.4	-1.2	2.4	-1.2	2.4	V
T1,T2	PSTA	XX	0	0	400	0	400	0	400	mV

- See footnotes on next page.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/541-1 for class I and M28787/541-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. Unless otherwise specified herein, the module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.5 V dc.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

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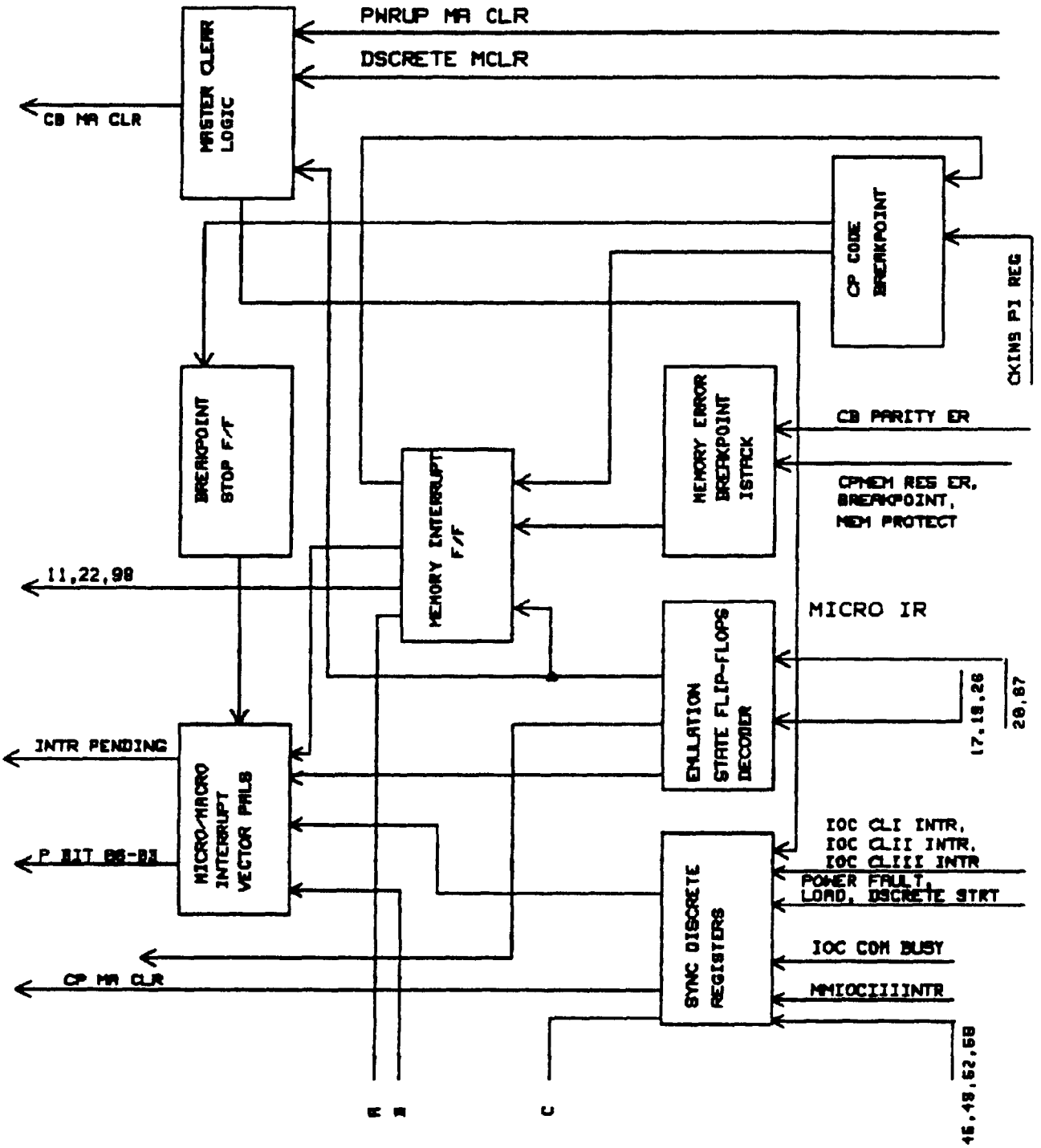


FIGURE 1. FWA functional block diagram.

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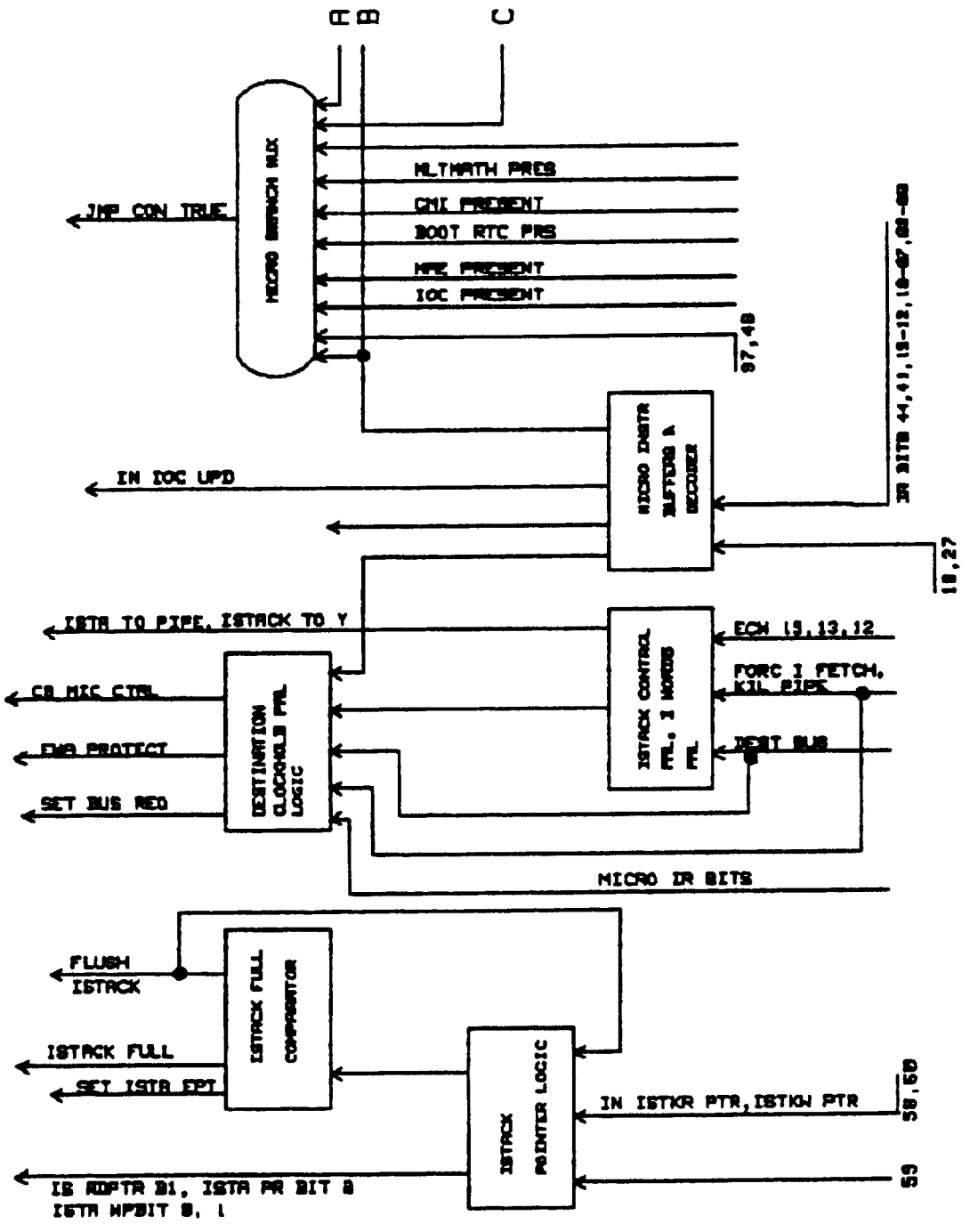


FIGURE 1. FWA functional block diagram - Continued.

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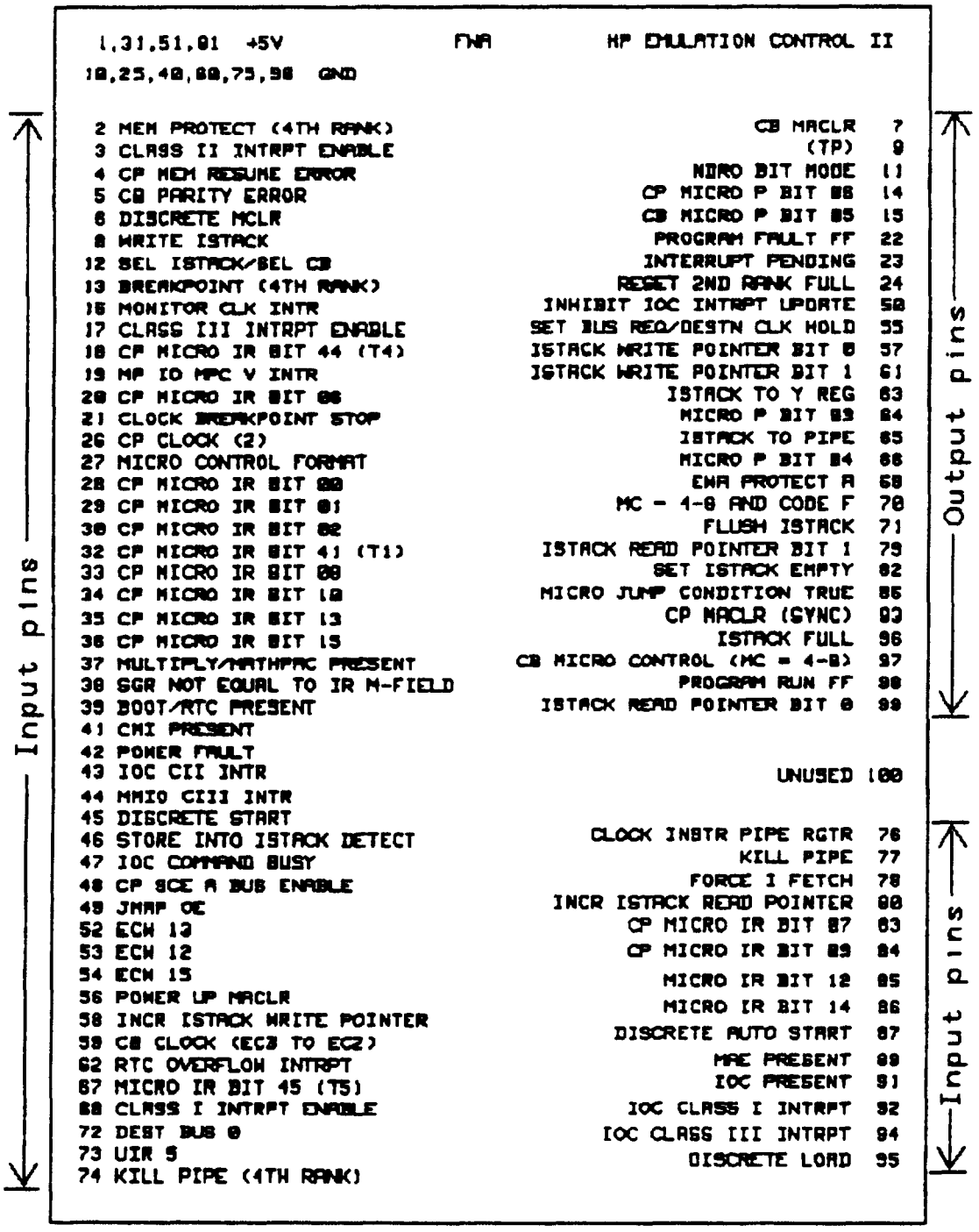


FIGURE 2. FWA pin assignment.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. Test conditions A, C, D, E, L and S are stress tests and will be performed only at 25 ± 5°C. If test conditions A, C, D, E, L or S are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E, L and S where applicable, followed by the remaining tests in the sequence.

4.4.3. Multiple testing of pins. Singular pins listed multiple times in a test table indicate multiple devices connected to the pins. The module shall be tested so that each device connected to a pin is exercised and tested.

4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition M, shall be performed using the initial limits at 25 ± 5°C ambient.

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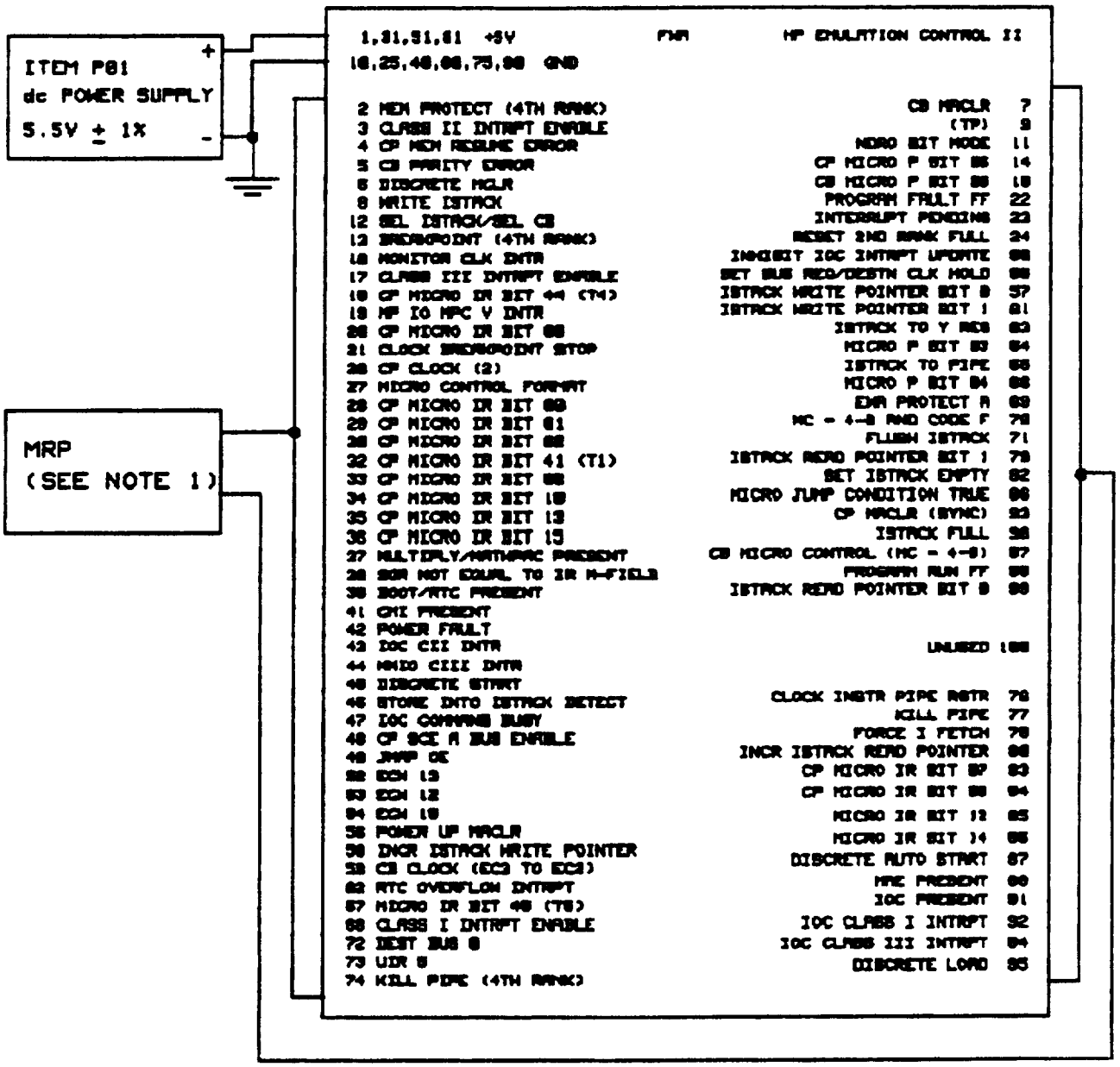
4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5. Following the life test and while still at the maximum class temperature, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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NOTE:

1. Military Reconfigurable Processor (MRP) or emulator capable of reproducing system backpanel characteristics.

FIGURE 3. FWA operating temperature and component temperature test circuit.

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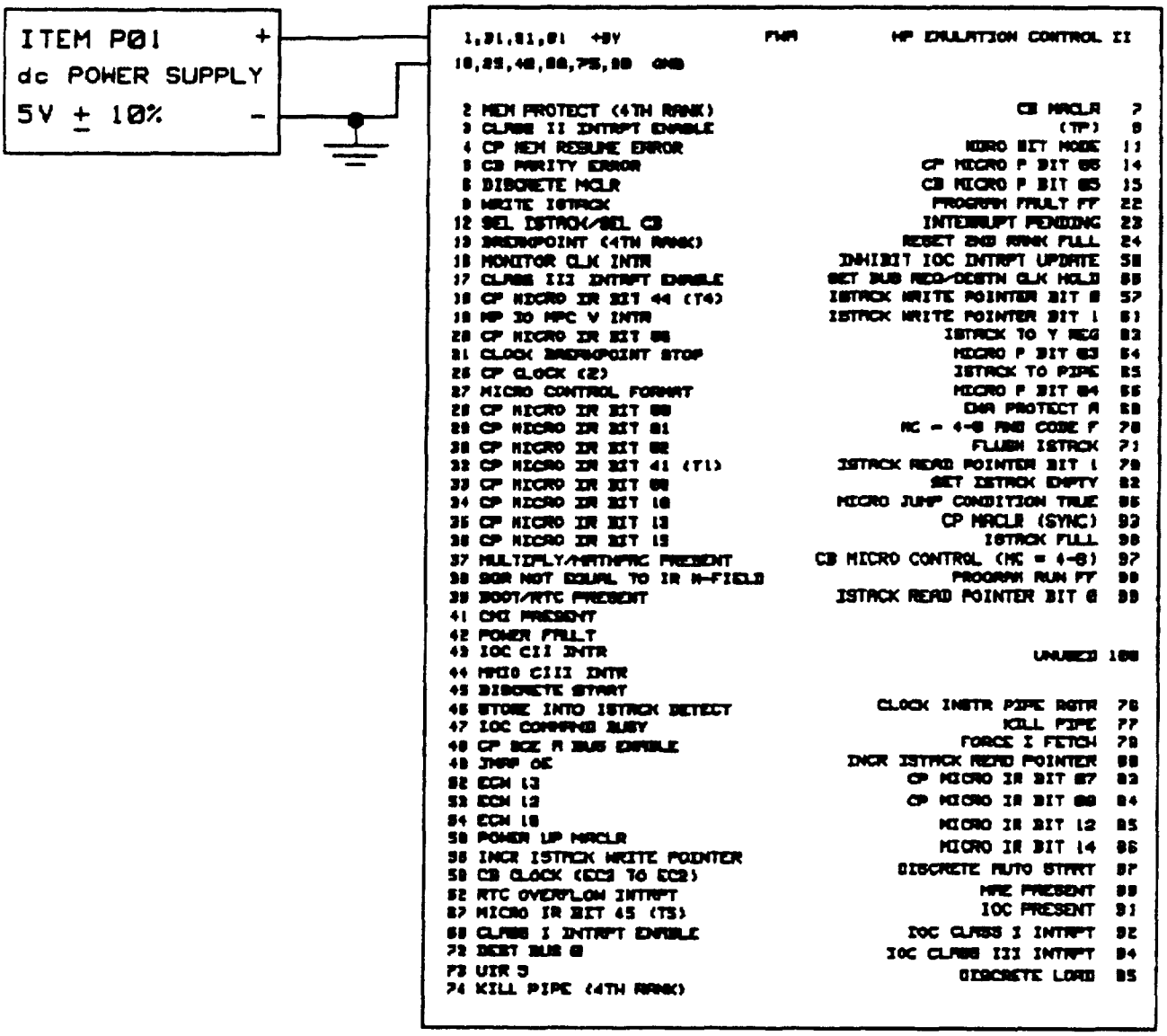
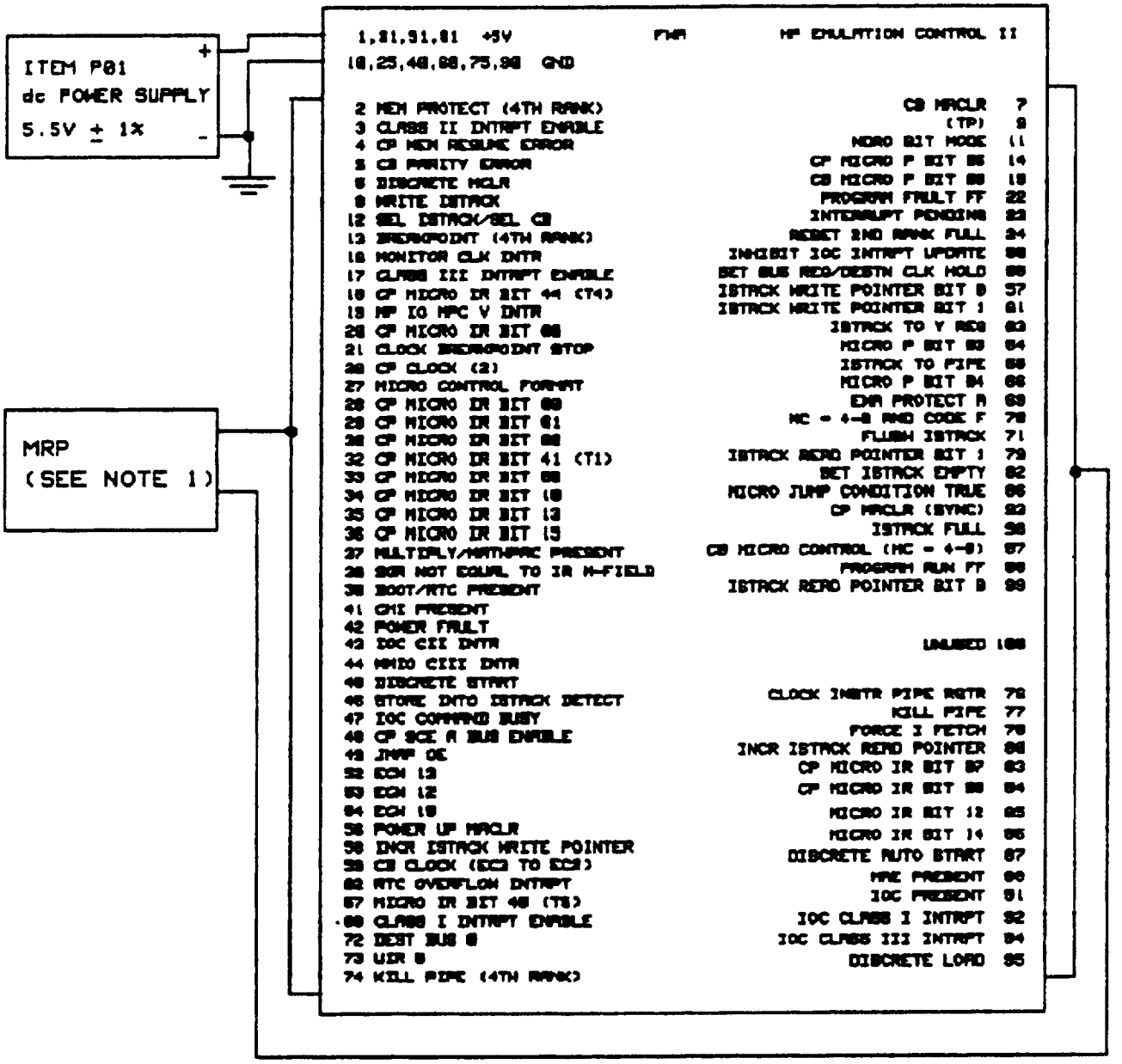


FIGURE 4. FWA vibration test circuit.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 5. FWA life test circuit.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Ibd = input breakdown current.
(c) Icc = power supply current.
(d) Iih = input high level current.
(e) Iil = input low level current.
(f) Iin = input forcing current.
(g) Ioh = output high level current.
(h) Iol = output low level current.
(i) Ios = output short circuit current.
(j) Iozh = output high level leakage current.
(k) Iozl = output low level leakage current.
(l) NA = not applicable.
(m) PSFC = power supply filter capacitance.
(n) PSTA = power supply transient amplitude.
(o) Vic = input clamp diode voltage.
(p) Vih = input high level voltage.
(q) Vil = input low level voltage.
(r) Vin = input forcing voltage.
(s) Voh = output high level voltage.
(t) Vol = output low level voltage.
(u) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the -55°C to 85°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 7.062 W (max); 5.457 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
MEM PROTECT	-1.2	0.4	2.4	-1.2	0.04	0.2	100
CLII INTR EN	-1.2	0.4	2.4	-1.1	0.07	2.1	125
CPMEM RES ER	-1.2	0.4	2.4	-0.6	0.02	0.1	50
CB PARITY ER	NA	0.4	2.4	-1.66	-0.54	0.12	125
DSCRETE MCLR	NA	0.4	2.4	-3.06	-0.51	1.02	125
WRITE ISTACK	-1.5	0.4	2.4	-0.8	0.04	0.1	75
SEL ISTK SEL	-1.2	0.4	2.4	-0.6	0.02	0.1	75
BREAKPOINT	-1.2	0.4	2.4	-1.0	0.04	0.2	75
MONITOR CLK	NA	0.4	2.4	-1.31	-0.54	1.02	100
CIII INTR EN	-1.5	0.4	2.4	-0.25	0.03	1.0	75
IR BIT 44	-1.5	0.4	2.4	-1.7	0.09	2.2	125
MPIOMPC VINT	-1.5	0.4	2.4	-0.25	0.03	1.0	75
IR BIT 06	-1.5	0.4	2.4	-0.4	0.02	0.1	75
CK BRKPT STP	-1.2	0.4	2.4	-2.6	0.07	1.1	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
CP CLOCK	-1.2	0.4	2.4	-5.0	0.14	2.2	125
MIC CTRL FMT	-1.2	0.4	2.4	-0.85	0.05	1.1	100
IR BIT 00	-1.2	0.4	2.4	-0.6	0.02	0.1	50
IR BIT 01	-1.2	0.4	2.4	-0.6	0.02	0.1	75
IR BIT 02	-1.2	0.4	2.4	-0.6	0.02	0.1	75
IR BIT 41	-1.5	0.4	2.4	-0.2	0.02	0.1	50
IR BIT 08	-1.2	0.4	2.4	-0.6	0.02	0.1	75
IR BIT 10	-1.2	0.4	2.4	-0.6	0.02	0.1	75
IR BIT 13	-1.5	0.4	2.4	-0.2	0.02	0.1	75
IR BIT 15	-1.2	0.4	2.4	-0.8	0.04	0.2	75
MLTMATH PRES	NA	0.4	2.4	-1.66	-0.54	0.12	75
SGR-EQ TO IR	-1.2	0.4	2.4	-0.6	0.02	0.1	75
BOOT RTC PRS	NA	0.4	2.4	-1.66	-0.54	0.12	50
CMI PRESENT	NA	0.4	2.4	-1.66	-0.54	0.12	50

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
POWER FAULT	NA	0.4	2.4	-1.26	-0.54	0.12	75
IOC CII INTR	NA	0.4	2.4	-1.26	-0.54	0.12	50
MMIOCIIIINTR	NA	0.4	2.4	-1.26	-0.54	0.12	75
DSCRETE STRT	NA	0.4	2.4	-1.26	-0.54	0.12	50
STOIN ISTK D	-1.5	0.4	2.4	-0.2	0.02	0.1	50
IOC COM BUSY	NA	0.4	2.4	-1.26	-0.54	0.12	75
CPSC ABUS EN	-1.2	0.4	2.4	-0.6	0.02	0.1	100
JMAP OE	-1.2	0.4	2.4	-0.6	0.02	0.1	100
ECW 13	-1.5	0.4	2.4	-0.85	0.05	1.1	100
ECW 12	-1.5	0.4	2.4	-0.85	0.05	1.1	100
ECW 15	-1.5	0.4	2.4	-0.85	0.05	1.1	100
PWRUP MA CLR	NA	0.4	2.4	-3.26	-0.49	1.12	175
IN ISTKW PTR	-1.2	0.4	2.4	-1.68	0.08	0.3	125
CB CLK	-1.2	0.4	2.4	-1.8	0.06	0.3	125

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
RTC OVFL INT	NA	0.4	2.4	-1.31	-0.54	1.02	125
IR BIT 45	-1.5	0.4	2.4	-0.85	0.05	1.1	125
CI INTR EN	-1.5	0.4	2.4	-0.85	0.05	1.1	100
DEST BUS	-1.5	0.4	2.4	-0.25	0.03	1.0	75
UIR 5	-1.5	0.4	2.4	-0.5	0.05	2.0	75
KILL PIPE	-1.2	0.4	2.4	-0.6	0.02	0.1	75
CKINS PI REG	-1.2	0.4	2.4	-2.6	0.07	1.1	75
KIL PIPE	-1.2	0.4	2.4	-2.05	0.09	1.3	100
FORC I FETCH	-1.2	0.4	2.4	-0.85	0.05	1.1	75
IN ISTKR PTR	-1.5	0.4	2.4	-2.28	0.1	0.4	125
IR BIT 07	-1.2	0.4	2.4	-0.6	0.02	0.1	50
IR BIT 09	-1.2	0.4	2.4	-0.6	0.02	0.1	50
IR BIT 12	-1.5	0.4	2.4	-0.2	0.02	0.1	75
IR BIT 14	-1.5	0.4	2.4	-0.8	0.04	0.2	100

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
DSCRT AUTOST	NA	0.4	2.4	-1.66	-0.54	0.12	75
MAE PRESENT	NA	0.4	2.4	-1.66	-0.54	0.12	75
IOC PRESENT	NA	0.4	2.4	-1.66	-0.54	0.12	50
IOC CI INTR	NA	0.4	2.4	-1.26	-0.54	0.12	75
IOC CIII INT	NA	0.4	2.4	-1.26	-0.54	0.12	75
LOAD	NA	0.4	2.4	-1.26	-0.54	0.12	75

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)
CB MA CLR	2.4	0.5	-0.4	6.0	NA	NA	-60.0
TP	2.4	0.5	-0.4	6.0	NA	NA	-40.0
NDRO BIT MOD	2.4	0.5	-0.16	1.6	NA	NA	-15.0
P BIT 06	2.4	0.5	-0.4	6.0	0.05	0.05	-60.0
P BIT 05	2.4	0.5	-0.4	6.0	0.05	0.05	-60.0
PRG FAULT FF	2.4	0.5	-0.16	1.6	NA	NA	-15.0
INTR PENDING	2.4	0.5	-0.16	1.6	NA	NA	-30.0
RST2 RNK FUL	2.4	0.5	-0.16	1.6	NA	NA	-15.0
IN IOC UPD	2.4	0.5	-0.4	6.0	NA	NA	-60.0
SET BUS REQ	2.4	0.5	-0.4	6.0	NA	NA	-60.0
ISTA WPBIT 0	2.4	0.5	-0.4	6.0	NA	NA	-60.0
ISTA WPBIT 1	2.4	0.5	-0.4	6.0	NA	NA	-60.0
ISTACK TO Y	2.4	0.5	-0.16	1.6	NA	NA	-30.0
P BIT 03	2.4	0.5	-0.4	6.0	0.05	0.05	-60.0

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)
ISTA TO PIPE	2.4	0.5	-0.4	6.0	NA	NA	-40.0
P BIT 04	2.4	0.5	-0.4	6.0	0.05	0.05	-60.0
EWA PROTECT	2.4	0.5	-0.16	1.6	NA	NA	-30.0
AND CODE F	2.4	0.5	-0.16	1.6	NA	NA	-30.0
FLUSH ISTACK	2.4	0.5	-0.4	6.0	NA	NA	-60.0
IS RDPTR B1	2.4	0.5	-0.4	6.0	NA	NA	-60.0
SET ISTA EPT	2.4	0.5	-0.16	1.6	NA	NA	-15.0
JMP CON TRUE	2.4	0.5	-0.4	6.0	0.05	0.05	-60.0
CP MA CLR	2.4	0.5	-0.16	1.6	NA	NA	-30.0
ISTACK FULL	2.4	0.5	-0.4	6.0	NA	NA	-60.0
CB MIC CTRL	2.4	0.5	-0.4	6.0	NA	NA	-60.0
PROG RUN FF	2.4	0.5	-0.4	6.0	NA	NA	-40.0
ISTA RPBIT 0	2.4	0.5	-0.4	6.0	NA	NA	-60.0

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6.3.3 Class I and II dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	CP CLK to NDRO BIT MOD	-	-	-	72	72	72	nS	12
1	tpd	CP CLK to PROG RUN FF	-	-	-	72	72	72	nS	12
2	tpd	CP CLK to RST2 RNK FUL	-	-	-	52	52	52	nS	12
3	tpd	CP CLK to IN IOC UPD	-	-	-	25	25	25	nS	12
4	tpd	SGR-EQ TO IR to JMP CON TRUE	-	-	-	16	16	16	nS	12
5	tpd	CP CLK to CP MA CLR	-	-	-	29	29	29	nS	12
6	tpd	PWRUP MA CLR to CB MA CLR	-	-	-	22	22	22	nS	12
6	tpd	DSCRETE MCLR to CB MA CLR	-	-	-	22	22	22	nS	12
7	tpd	IR BIT 44 to INTR PENDING	-	-	-	90	90	90	nS	12
8	tpd	CK BRKPT STP to INTR PENDING	-	-	-	71	71	71	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
9	tpd	IR BIT 44 to P BIT 06	-	-	-	101	101	101	nS	12
9	tpd	IR BIT 44 to P BIT 05	-	-	-	101	101	101	nS	12
9	tpd	IR BIT 44 to P BIT 03	-	-	-	101	101	101	nS	12
9	tpd	IR BIT 44 to P BIT 04	-	-	-	101	101	101	nS	12
10	tpd	CP CLK to SET BUS REQ	-	-	-	92	92	92	nS	12
11	tpd	MIC CTRL FMT to CB MIC CTRL	-	-	-	36	36	36	nS	12
12	tpd	IR BIT 09 to EWA PROTECT	-	-	-	58	58	58	nS	12
13	tpd	CP CLK to ISTACK TO Y	-	-	-	92	92	92	nS	12
14	tpd	KILL PIPE to FLUSH ISTACK	-	-	-	13	13	13	nS	12
15	tpd	CP CLK to FLUSH ISTACK	-	-	-	26	26	26	nS	12
16	tpd	CB CLK to ISTACK FULL	-	-	-	15	15	15	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
17	tpd	CB CLK to ISTACK WPBIT 1	-	-	-	13	13	13	nS	12
17	tpd	CB CLK to ISTACK WPBIT 0	-	-	-	13	13	13	nS	12
17	tpd	CB CLK to IS RDPTR B1	-	-	-	13	13	13	nS	12
17	tpd	CB CLK to ISTA RPBIT 0	-	-	-	13	13	13	nS	12
18	tpd	CB CLK to SET ISTA EPT	-	-	-	45	45	45	nS	12
19	tpd	IN ISTKR PTR to SET ISTA EPT	-	-	-	22	22	22	nS	12
19	tpd	IN ISTKW PTR to SET ISTA EPT	-	-	-	22	22	22	nS	12
20	tsu	MEM PRO-TECT to CKINS PI REG	37	37	37	-	-	-	nS	12
20	tsu	CLII INTR EN to CKINS PI REG	37	37	37	-	-	-	nS	12
20	tsu	CPMEM RES ER to CKINS PI REG	37	37	37	-	-	-	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
20	tsu	CB PARITY ER to CKINS PI REG	37	37	37	-	-	-	nS	12
21	th	CKINS PI REG to MEM PRO-TECT	7	7	7	-	-	-	nS	12
21	th	CKINS PI REG to CLII INTR EN	7	7	7	-	-	-	nS	12
21	th	CKINS PI REG to CPMEM RES ER	7	7	7	-	-	-	nS	12
21	th	CKINS PI REG to CB PARITY ER	7	7	7	-	-	-	nS	12
22	tpd	CPSC ABUS EN to JMP CON TRUE	-	-	-	28	28	28	nS	12
23	tsu	POWER FAULT to CP CLOCK	7	7	7	-	-	-	nS	12
23	tsu	IOC CII INTR to CP CLOCK	7	7	7	-	-	-	nS	12
23	tsu	MMIOCIII-INTR to CP CLOCK	7	7	7	-	-	-	nS	12
23	tsu	DSCRETE STRT to CP CLOCK	7	7	7	-	-	-	nS	12
23	tsu	IOC COM BUSY to CP CLOCK	7	7	7	-	-	-	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
23	tsu	IOC CI INTR to CP CLOCK	7	7	7	-	-	-	nS	12
23	tsu	IOC CIII INT to CP CLOCK	7	7	7	-	-	-	nS	12
23	tsu	LOAD to CP CLOCK	7	7	7	-	-	-	nS	12
24	th	CP CLOCK to POWER FAULT	14	14	14	-	-	-	nS	12
24	th	CP CLOCK to IOC CII INTR	14	14	14	-	-	-	nS	12
24	th	CP CLOCK to MMIOC-IIIINTR	14	14	14	-	-	-	nS	12
24	th	CP CLOCK to DSCRETE STRT	14	14	14	-	-	-	nS	12
24	th	CP CLOCK to IOC COM BUSY	14	14	14	-	-	-	nS	12
24	th	CP CLOCK to IOC CI INTR	14	14	14	-	-	-	nS	12
24	th	CP CLOCK to IOC CIII INT	14	14	14	-	-	-	nS	12
24	th	CP CLOCK to LOAD	14	14	14	-	-	-	nS	12
25	tsu	PWRUP MA CLR to CP CLOCK	29	29	29	-	-	-	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
26	th	CP CLOCK to PWRUP MA CLR	9	9	9	-	-	-	nS	12
27	te	JMAP OE to P BIT 06	-	-	-	27	27	27	nS	12
27	te	JMAP OE to P BIT 05	-	-	-	27	27	27	nS	12
27	te	JMAP OE to P BIT 03	-	-	-	27	27	27	nS	12
27	te	JMAP OE to P BIT 04	-	-	-	27	27	27	nS	12
28	tsu	ECW 13 to CP CLOCK	65	65	65	-	-	-	nS	12
28	tsu	ECW 12 to CP CLOCK	65	65	65	-	-	-	nS	12
28	tsu	ECW 15 to CP CLOCK	65	65	65	-	-	-	nS	12
28	tsu	KIL PIPE to CP CLOCK	65	65	65	-	-	-	nS	12
29	th	CP CLOCK to ECW 13	10	10	10	-	-	-	nS	12
29	th	CP CLOCK to ECW 12	10	10	10	-	-	-	nS	12
29	th	CP CLOCK to ECW 15	10	10	10	-	-	-	nS	12
29	th	CP CLOCK to KIL PIPE	10	10	10	-	-	-	nS	12
29	th	CP CLOCK to FORC I FETCH	10	10	10	-	-	-	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
30	tsu	DEST BUS to CP CLOCK	42	42	42	-	-	-	nS	12
31	th	CP CLK to DEST BUS	10	10	10	-	-	-	nS	12
32	tsu	IN ISTKR PTR to CB CLOCK	24	24	24	-	-	-	nS	12
32	tsu	IN ISTKW PTR to CB CLOCK	24	24	24	-	-	-	nS	12
33	th	CB CLOCK to IN ISTKR PTR	1	1	1	-	-	-	nS	12
33	th	CB CLOCK to IN ISTKW PTR	1	1	1	-	-	-	nS	12

6.4 Module function. Emulation control provides the control logic to emulate the macroinstructions, process interrupts, communicate over the common bus, and master clear the MRP. High performance emulation control is contained on three modules: Emulation Control 1 (EWAH), 2 (FWA), and 3 (FWB). Macroinstructions are loaded into the pipe register from memory via the common bus. For double length instructions, the Y word is loaded into the Y register. Macroinstruction emulation is performed by executing a series of short microprogram subroutines stored in micro memory. The subroutine starting address is contained in an Emulation Control Word (ECW) stored in the ECW Programmable Read-Only Memory (PROM). The ECW contains control bits along with the address pointer. A group of eight words are associated with each macroinstruction. The group is addressed by the macroinstruction operation (Op) and format codes (bits 15-8). An ECW within the group is addressed by address control. Address control determines if indexing and/or indirect addressing is specified, if the instruction is being executed remotely, and what interim sequences are required to execute the instruction.

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Status register 2 (bits 15-18) holds the indirect addressing information. Status register 1 holds Central Processor (CP) status bits that provide interrupt class lockouts, results of arithmetic operations, selection of page register sets, general register sets, Nondestructive Read Out (NDRO) memory, and CP mode selection (executive mode or task mode). The executive mode has a separate set of privileged instructions that cannot be executed in the task mode (an attempt to execute a privileged instruction in the task mode will result in a class II interrupt). Executive mode instructions are flagged. Conditional jump instructions (Op code = 40 (base 8)) are interpreted by conditional jump/stop control to determine if the specified jump condition is true. If true, the next macroinstruction is fetched from the jump address memory location. Interrupt control detects interrupts, determines the interrupt with the highest priority, and generates a 4-bit interrupt pointer. The interrupt pointer is used to form the starting address of the microprogram subroutine that processes the interrupt. Bus master control provides the CP control interface with the common bus. The CP is always the bus master when communicating with the Input/Output Controllers (IOCs) or memory interfaces. The bus is requested from the bus arbitrator and when the request is granted, bus master control generates a 4-bit function code and the control signals to perform the transaction. A MRP master clear is initiated when the MRP is powered-up or by the maintenance processor. Master clear places the MRP in the following initial condition:

(a) P register cleared.

(b) Status registers 1 and 2 cleared.

(c) Real Time Clock (RTC) and monitor clock functions disabled.

(d) Page registers are set to their own register address.

(e) All I/O channels cleared as follows:

1. Status words cleared.
2. All data buffers and chains deactivated.
3. All interrrupt data storage disabled, all EIEs cleared.
4. Class III interrupts disabled.
5. Serial monitor and suppress flags cleared.
6. Intercomputer time out function cleared.

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(f) Bus initialization signal generated.

(g) Run mode selected (only occurs on power-up master clear).

(h) Program and power fault cleared.

6.4.1 Microinstruction buffers. The microinstruction buffers receive the microinstruction microcontrol fields (bits 15-0). These bits are sent to the destination clock hold logic, the microbranch multiplexer, and the micro/macro interrupt Programmable Array Logic (PALs). The microinstruction buffers are constantly enabled.

6.4.2 Instruction Stack (ISTACK) controls. ISTACK controls are located on emulation control 2 and 3 modules. The following controls are located on the emulation control 2 module:

(a) ISTACK full.

(b) Flush stack.

(c) ISTACK empty.

(d) ISTACK pointer logic.

The ISTACK full flip-flop is set when the write pointer is three counts ahead of the read pointer, and a word is loaded into the ISTACK. The increment ISTACK read pointer signal causes the flip-flop to reset (in other words, a word is read from the ISTACK). The ISTACK full signal is used to disable the clock ISTACK logic until the top location in the stack is empty. A stack flush condition is caused by the execution of a kill pipe microinstruction. When this occurs:

(a) Pipe full is reset.

(b) ISTACK read and write pointers are reset to zero.

(c) ISTACK empty signal is set.

(d) ISTACK ready signal is reset.

(e) ISTACK full signal is reset.

(f) Clock pipe/Y and breakpoint stop are inhibited.

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After killing the pipe, an instruction fetch is executed to reload the ISTACK (instruction to Y and instruction to pipe are initialized). The capability to do instruction fetches is contingent upon the flush stack flip-flop being reset. The flush stack flip-flop is reset at clock 4th rank time by the kill pipe 4R signal or a master clear. ISTACK empty logic is composed of a comparator that checks if the write pointer is one count ahead of the read pointer (for example, ISTACK empty is set when the last word is read out of the ISTACK). The ISTACK empty signal is reset when a word is again written into the ISTACK. The ISTACK Read Pointer (RPTR) and Write Pointer (WPTR) are composed of two 4-bit shift registers that increment after each instruction word is written into the ISTACK or read from the ISTACK. In the ISTACK write sequence, the RPTR remains constant; in the ISTACK read sequence, the WPTR remains constant. Note that the address sequence is 0, 2, 3, 1. The flush stack signal causes both shift registers to be reset to zero.

6.4.3 Macro/micro interrupt circuits. The macro/micro interrupt circuits consist of the micro/macro interrupt PALs, the memory interrupt flip-flop, the sync registers, memory error logic, the Op code/breakpoint stop flip-flop, the microcontrol flip-flops, and a microbranch multiplexer. The micro/macro interrupt PALs detect interrupt inputs, determine their priority, and generate a 4-bit code. This 4-bit code from the PAL multiplexer is inserted into micro P bus bit 6-3 to form the starting address of the micro-program interrupt routine when an emulate start microinstruction is executed (P = 2, and microinstruction bit 41 = 0 at the start interrupt decoder). The sync register receives IOC, Memory Mapped Input/Output (MMIO), and power fault macro interrupts along with emulate start, ISTACK, load, and start micro interrupts and sync them to the phase normal clock. Outputs are sent to the interrupt macro/micro control PALs and micro branch multiplexer. The micro branch multiplexer determines if a branch condition is true based upon the microinstruction BC field (bits 15-12 of a format 1 microinstruction). Breakpoint logic consists of the Op code breakpoint flip-flop and the breakpoint stop flip-flop. Breakpoint matches on an instruction fetch are saved in the Op code breakpoint flip-flop at clock pipe time and then transferred to the breakpoint stop flip-flop at instruction execution time to ensure the CP does not stop before executing the macroinstruction. A breakpoint match on the second word of a macroinstruction is loaded into the breakpoint stop flip-flop when the second word is loaded into the Y register (ISTACK to Y or CBRDATA to Y). Breakpoint matches on operand references are loaded into the breakpoint stop flip-flop at Data Terminator

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Ready (DTR) time. The output from the breakpoint stop flip-flop is sent to the macro/micro interrupt PALs for interrupt vector coding. Memory interrupt logic is composed of the memory error logic and the memory error flip-flop. The memory error logic includes a 4 word by 4-bit file to effectively extend the ISTACK by 4-bits. This file holds memory errors associated with a given word in the ISTACK. Additionally, memory errors may be selected from the common bus at the memory error multiplexer. Outputs from the memory error multiplexer are sent to their respective memory error flip-flop and loaded at clock pipe time. Outputs from the memory error flip-flops are sent to the macro/micro interrupt PALs for interrupt vector coding.

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6.4.4 Description of I/O pins.

Signal name	Pin number	I/O	Description
MEM PROTECT	2	I	MEM PROTECT (4TH RANK)
CLII INTR EN	3	I	CLASS II INTRPT ENABLE
CPMEM RES ER	4	I	CP MEMORY RESUME ERROR
CB PARITY ER	5	I	CB PARITY ERROR
DSCRETE MCLR	6	I	DISCRETE MCLR
WRITE ISTACK	8	I	WRITE ISTACK
SEL ISTK SEL	12	I	SEL ISTACK/SEL CB
BREAKPOINT	13	I	BREAKPOINT (4TH RANK)
MONITOR CLK	16	I	MONITOR CLK INTR
CIII INTR EN	17	I	CLASS III INTRPT ENABLE
IR BIT 44	18	I	CP MICRO IR BIT 44 (T4)
MPIOMPC VINT	19	I	MP IO MPC V INTR
IR BIT 06	20	I	CP MICRO IR BIT 06
CK BRKPT STP	21	I	CLOCK BREAKPOINT STOP
CP CLOCK	26	I	CP CLOCK (2)
MIC CTRL FMT	27	I	MICRO CONTROL FORMAT
IR BIT 00	28	I	CP MICRO IR BIT 00
IR BIT 01	29	I	CP MICRO IR BIT 01
IR BIT 02	30	I	CP MICRO IR BIT 02
IR BIT 41	32	I	CP MICRO IR BIT 41 (T1)

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6.4.4 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
IR BIT 08	33	I	CP MICRO IR BIT 08
IR BIT 10	34	I	CP MICRO IR BIT 10
IR BIT 13	35	I	CP MICRO IR BIT 13
IR BIT 15	36	I	CP MICRO IR BIT 15
MLTMATH PRES	37	I	MULTIPLY/MATHPAC PRESENT
SGR-EQ TO IR	38	I	SGR NOT EQUAL TO IR M-FIELD
BOOT RTC PRS	39	I	BOOT RTC PRESENT
CMI PRESENT	41	I	CMI PRESENT
POWER FAULT	42	I	POWER FAULT
IOC CII INTR	43	I	IOC CII INTR
MMIOCIIIINTR	44	I	MMIO CIII INTR
DSCRETE STRT	45	I	DISCRETE START
STOIN ISTK D	46	I	STORE INTO ISTACK DETECT
IOC COM BUSY	47	I	IOC COMMAND BUSY
CPSC ABUS EN	48	I	CP SCE A BUS ENABLE
JMAP OE	49	I	JMAP OE
ECW 13	52	I	ECW 13
ECW 12	53	I	ECW 12
ECW 15	54	I	ECW 15
PWRUP MA CLR	56	I	PWR UP MACLR

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6.4.4 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
IN ISTKW PTR	58	I	INCREMENT ISTACK WRITE POINTER
CB CLK	59	I	CB CLK (EC3 TO EC2)
RTC OVFL INT	62	I	RTC OVERFLOW INTR
IR BIT 45	67	I	MICRO IR BIT 45 (T5)
CI INTR EN	68	I	CLASS I INTR ENABLE
DEST BUS	72	I	DEST BUS 0
UIR 5	73	I	UIR 5
KILL PIPE	74	I	KILL PIPE (4TH RANK)
CKINS PI REG	76	I	CLOCK INSTR PIPE RGTR
KIL PIPE	77	I	KILL PIPE
FORC I FETCH	78	I	FORCE I FETCH
IN ISTKR PTR	80	I	INCR ISTACK READ POINTER
IR BIT 07	83	I	CP MICRO IR BIT 07
IR BIT 09	84	I	CP MICRO IR BIT 09
IR BIT 12	85	I	CP MICRO IR BIT 12
IR BIT 14	86	I	CP MICRO IR BIT 14
DSCRT AUTOST	87	I	DISCRETE AUTO START
MAE PRESENT	89	I	MAE PRESENT
IOC PRESENT	91	I	IOC PRESENT
IOC CI INTR	92	I	IOC CLASS I INTRPT

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6.4.4 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
IOC CIII INT	94	I	IOC CLSS III INTRPT
LOAD	95	I	DISCRETE LOAD
CB MA CLR	7	O	CB MACLR
TP	9	O	TP
NDRO BIT MOD	11	O	NDRO BIT MODE
P BIT 06	14	O	CP MICRO P BIT 06
P BIT 05	15	O	CP MICRO P BIT 05
PRG FAULT FF	22	O	PROGRAM FAULT FF
INTR PENDING	23	O	INTERRUPT PENDING
RST2 RNK FUL	24	O	RESET 2ND RANK FULL
IN IOC UPD	50	O	INHIBIT IOC INTRPT UPDATE
SET BUS REQ	55	O	SET BUS REQ/DESTN CLK HOLD
ISTA WPBIT 0	57	O	ISTACK WRITE POINTER BIT 0
ISTA WPBIT 1	61	O	ISTACK WRITE POINTER BIT 1
ISTACK TO Y	63	O	ISTACK TO Y REG
P BIT 03	64	O	MICRO P BIT 03
ISTA TO PIPE	65	O	ISTACK TO PIPE
P BIT 04	66	O	MICRO P BIT 04
EWA PROTECT	69	O	EWA PROTECT A
AND CODE F	70	O	MC = 4-8 AND CODE F

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6.4.4 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
FLUSH ISTACK	71	O	FLUSH ISTACK
IS RDPTR B1	79	O	ISTACK READ POINTER BIT 1
SET ISTA EPT	82	O	SET ISTACK EMPTY
JMP CON TRUE	88	O	MICRO JUMP CONDITION TRUE
CP MA CLR	93	O	CP MACLR (SYNC)
ISTACK FULL	96	O	ISTACK FULL
CB MIC CTRL	97	O	CB MICRO CONTROL (MC = 4-8)
PROG RUN FF	98	O	PROGRAM RUN FF
ISTA RPBIT 0	99	O	ISTACK READ POINTER BIT 0

6.5 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Preparing activity:
Navy - SH

(Project 5963-N712)

Agent:
NWSC Crane

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TABLE II. Isolation test.

Test cond	Test description	Connect A to	Connect B to	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A1	Isolation, frame	C,100	Frame	10	INF	MOhm	1-8
A2	Isolation, unused to used pins	100	C	10	INF	MOhm	1-8

NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A1,A2
Isolation measurement	NA	± 500 KOhm	A1,A2

2. The ohmmeter used shall be item R06.

3. In the set-up conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, and 'B' shall correspond to the negative terminal and 'C' shall correspond to all used pins shorted together.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

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NOTES: - Continued.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

6. Caution: Set megohm bridge to limit current to 10 uA before applying voltage to the module.

7. The test circuit shall be as indicated on figure 6.

8. Abbreviations used are defined in 6.1.

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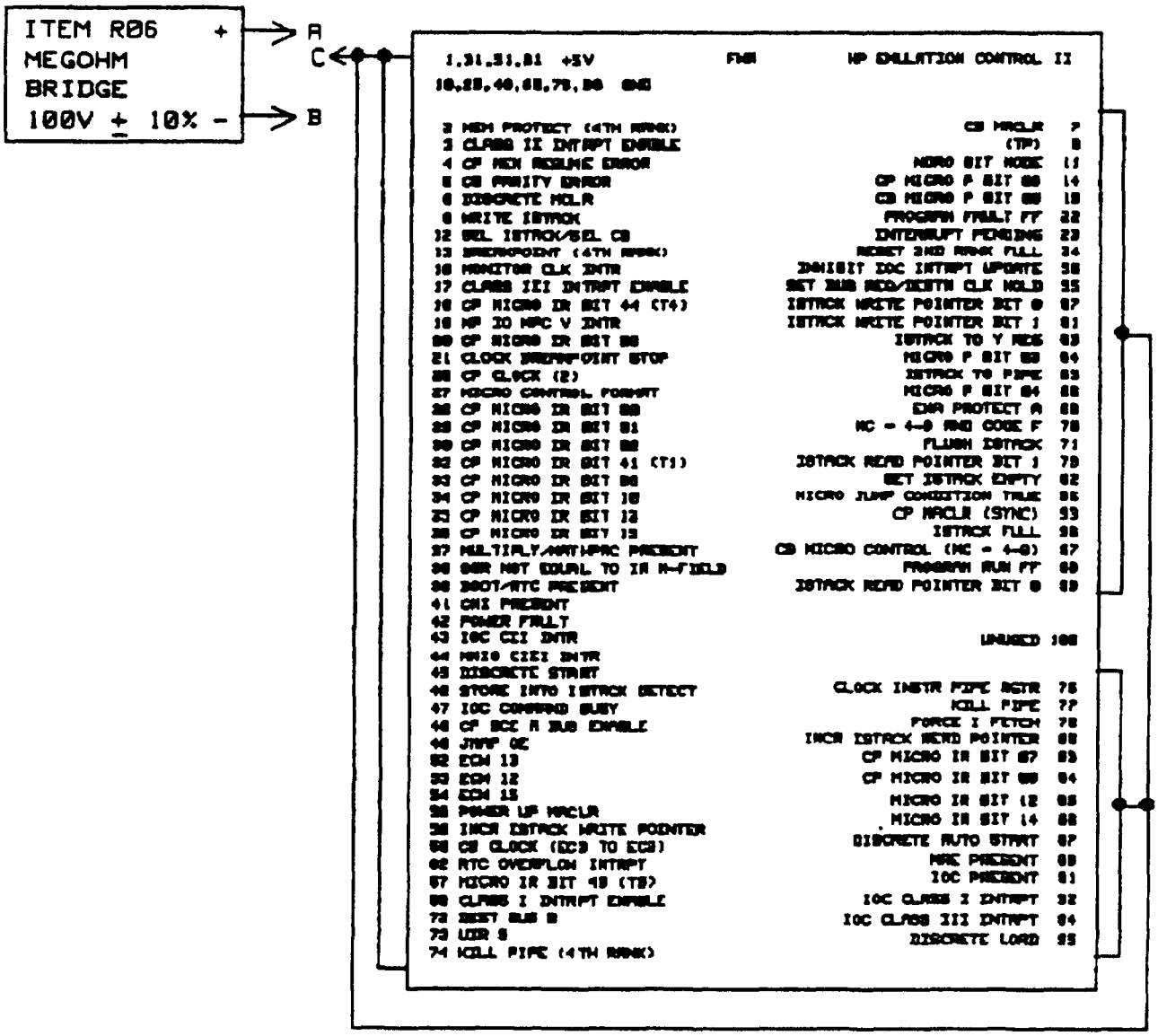


FIGURE 6. Isolation test circuit for table II.

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TABLE III. Continuity test.

Test cond	Test description	Connect		25±5°C Initial limits		Units	Notes
		A to	B to	Min	Max		
B1	Continuity, GND	10	25	0	1	Ohm	1-5
B2	Continuity, GND	10	40				
B3	Continuity, GND	60	75				
B4	Continuity, GND	60	90				
B5	Continuity, +5V	1	31				
B6	Continuity, +5V	51	81	0	1	Ohm	1-5

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA	± 0.05 Ohm	B1-B6

2. The ohmmeter used shall be item R07.

3. In the set-up conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. The test circuit shall be as indicated on figure 7.

5. Abbreviations used are defined in 6.1.

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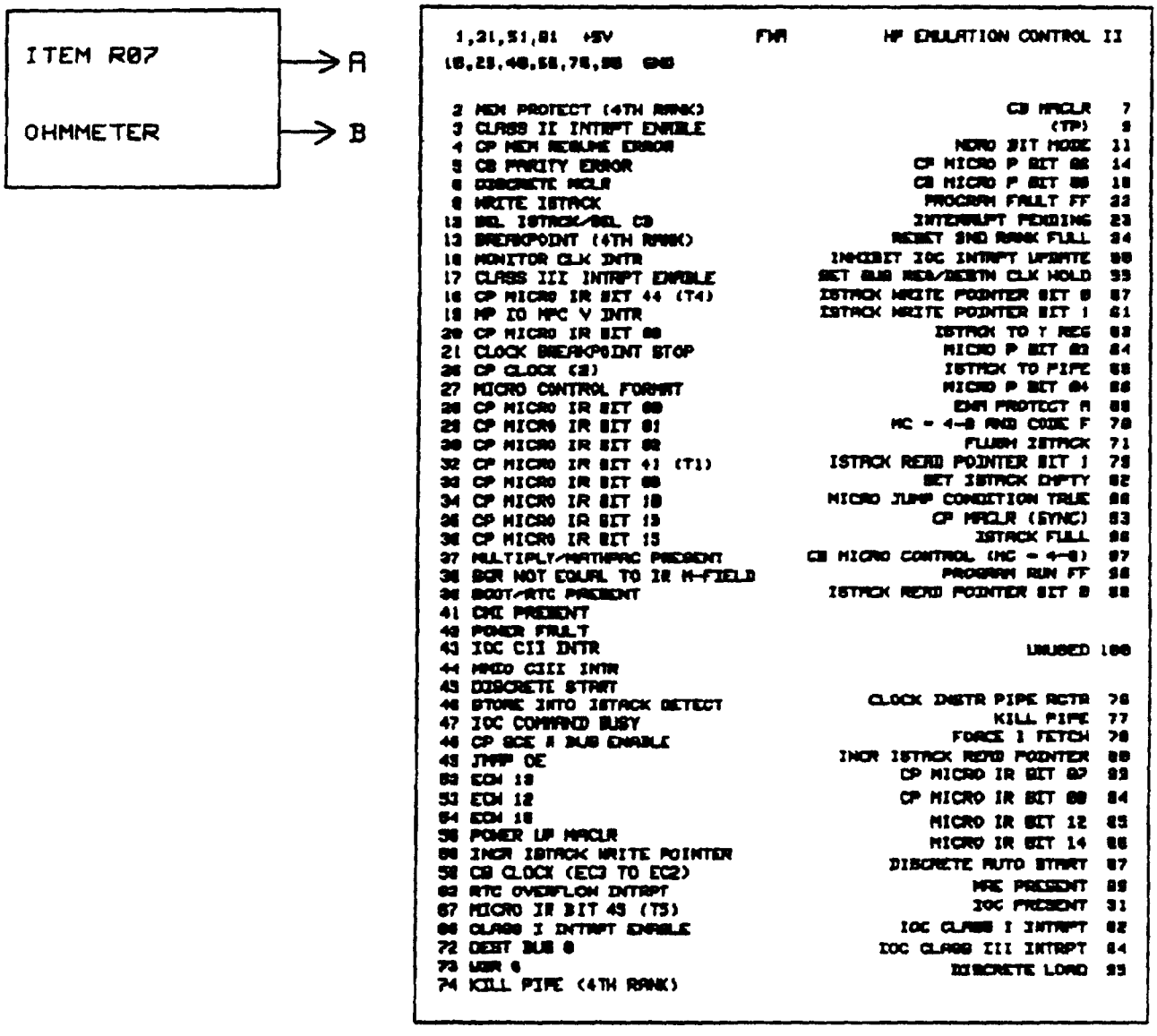


FIGURE 7. Continuity test circuit for table III.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect		25 ± 5°C Initial limits	Notes
			Positive lead to	Negative lead to		
C	Vcc	7.0	1,31,51, 81	10,25, 40,60, 75,90	Module shall pass all remaining tests.	1-6

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV,-70 mV	C
Power supply current limit	2.31 A	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. The test circuit shall be as indicated on figure 8.

6. Abbreviations used are defined in 6.1.

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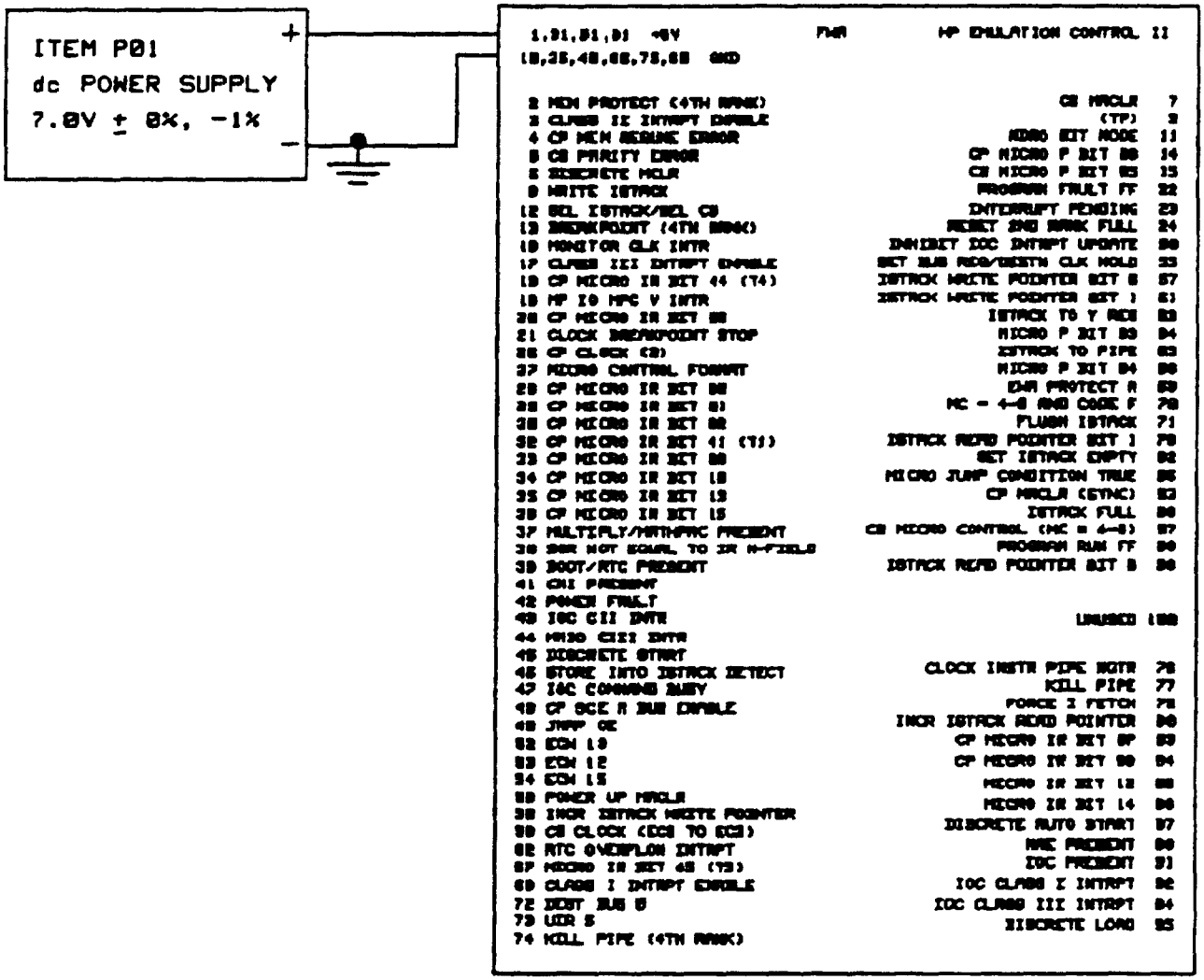


FIGURE 8. Vcc stress test circuit for table IV.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D1	MEM PROTECT	2	5.5	0.0	0.2	mA	1-3
D2	CLII INTR EN	3	5.5	0.0	2.1	mA	1-3
D3	CPMEM RES ER	4	5.5	0.0	0.1	mA	1-3
D4	CB PARITY ER	5	5.5	-0.02	0.12	mA	1-3
D5	DSCRETE MCLR	6	5.5	-0.02	1.02	mA	1-3
D6	WRITE ISTACK	8	5.5	0.0	0.1	mA	1-3
D7	SEL ISTK SEL	12	5.5	0.0	0.1	mA	1-3
D8	BREAKPOINT	13	5.5	0.0	0.2	mA	1-3
D9	MONITOR CLK	16	5.5	-0.02	1.02	mA	1-3
D10	CIII INTR EN	17	5.5	0.0	1.0	mA	1-3
D11	IR BIT 44	18	5.5	0.0	2.2	mA	1-3
D12	MPIOMPC VINT	19	5.5	0.0	1.0	mA	1-3
D13	IR BIT 06	20	5.5	0.0	0.1	mA	1-3
D14	CK BRKPT STP	21	5.5	0.0	1.1	mA	1-3
D15	CP CLOCK	26	5.5	0.0	2.2	mA	1-3
D16	MIC CTRL FMT	27	5.5	0.0	1.1	mA	1-3
D17	IR BIT 00	28	5.5	0.0	0.1	mA	1-3
D18	IR BIT 01	29					
D19	IR BIT 02	30					
D20	IR BIT 41	32	5.5	0.0	0.1	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D21	IR BIT 08	33	5.5	0.0	0.1	mA	1-3
D22	IR BIT 10	34					
D23	IR BIT 13	35	5.5	0.0	0.1	mA	1-3
D24	IR BIT 15	36	5.5	0.0	0.2	mA	1-3
D25	MLTMATH PRES	37	5.5	-0.02	0.12	mA	1-3
D26	SGR-EQ TO IR	38	5.5	0.0	0.1	mA	1-3
D27	BOOT RTC PRS	39	5.5	-0.02	0.12	mA	1-3
D28	CMI PRESENT	41					
D29	POWER FAULT	42					
D30	IOC CII INTR	43					
D31	MMIOCIIIINTR	44					
D32	DSCRETE STRT	45	5.5	-0.02	0.12	mA	1-3
D33	STOIN ISTK D	46	5.5	0.0	0.1	mA	1-3
D34	IOC COM BUSY	47	5.5	-0.02	0.12	mA	1-3
D35	CPSC ABUS EN	48	5.5	0.0	0.1	mA	1-3
D36	JMAP OE	49	5.5	0.0	0.1	mA	1-3
D37	ECW 13	52	5.5	0.0	1.1	mA	1-3
D38	ECW 12	53					
D39	ECW 15	54	5.5	0.0	1.1	mA	1-3
D40	PWRUP MA CLR	56	5.5	-0.02	1.12	mA	1-3
D41	IN ISTKW PTR	58	5.5	0.0	0.3	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D42	CB CLK	59	5.5	0.0	0.3	mA	1-3
D43	RTC OVFL INT	62	5.5	-0.02	1.02	mA	1-3
D44	IR BIT 45	67	5.5	0.0	1.1	mA	1-3
D45	CI INTR EN	68	5.5	0.0	1.1	mA	1-3
D46	DEST BUS	72	5.5	0.0	1.0	mA	1-3
D47	UIR 5	73	5.5	0.0	2.0	mA	1-3
D48	KILL PIPE	74	5.5	0.0	0.1	mA	1-3
D49	CKINS PI REG	76	5.5	0.0	1.1	mA	1-3
D50	KIL PIPE	77	5.5	0.0	1.3	mA	1-3
D51	FORC I FETCH	78	5.5	0.0	1.1	mA	1-3
D52	IN ISTKR PTR	80	5.5	0.0	0.4	mA	1-3
D53	IR BIT 07	83	5.5	0.0	0.1	mA	1-3
D54	IR BIT 09	84					
D55	IR BIT 12	85	5.5	0.0	0.1	mA	1-3
D56	IR BIT 14	86	5.5	0.0	0.2	mA	1-3
D57	DSCRT AUTOST	87	5.5	-0.02	0.12	mA	1-3
D58	MAE PRESENT	89					
D59	IOC PRESENT	91					
D60	IOC CI INTR	92					
D61	IOC CIII INT	94					
D62	LOAD	95	5.5	-0.02	0.12	mA	1-3

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NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	D1-D62
Power supply current limit	1.93 A	NA	D1-D62
Forcing voltage, (Vin)	5.5 V	± 55 mV	D1-D62
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 D1-D62 D1-D62
Input current measurement	NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA	± 10 uA ± 105 uA ± 5 uA ± 7 uA ± 52 uA ± 5 uA ± 10 uA ± 52 uA ± 50 uA ± 110 uA ± 50 uA ± 5 uA ± 55 uA ± 110 uA ± 55 uA ± 5 uA ± 10 uA ± 7 uA ± 5 uA ± 7 uA ± 5 uA ± 7 uA	D1 D2 D3 D4 D5 D6,D7 D8 D9 D10 D11 D12 D13 D14 D15 D16 D17-D23 D24 D25 D26 D27-D32 D33 D34

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input current measurement	NA	± 5 uA	D35,D36
	NA	± 55 uA	D37-D39
	NA	± 57 uA	D40
	NA	± 15 uA	D41,D42
	NA	± 52 uA	D43
	NA	± 55 uA	D44,D45
	NA	± 50 uA	D46
	NA	± 100 uA	D47
	NA	± 5 uA	D48
	NA	± 55 uA	D49
	NA	± 65 uA	D50
	NA	± 55 uA	D51
	NA	± 20 uA	D52
	NA	± 5 uA	D53-D55
	NA	± 10 uA	D56
	NA	± 7 uA	D57-D62

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E1	CB MA CLR	7	0.0	-150.0	-60.0	mA	1-4
E2	TP	9	0.0	-100.0	-40.0	mA	1-4
E3	NDRO BIT MOD	11	0.0	-100.0	-15.0	mA	1-4
E4	P BIT 06	14	0.0	-150.0	-60.0	mA	1-4
E5	P BIT 05	15	0.0	-150.0	-60.0	mA	1-4
E6	PRG FAULT FF	22	0.0	-100.0	-15.0	mA	1-4
E7	INTR PENDING	23	0.0	-130.0	-30.0	mA	1-4
E8	RST2 RNK FUL	24	0.0	-100.0	-15.0	mA	1-4
E9	IN IOC UPD	50	0.0	-150.0	-60.0	mA	1-4
E10	SET BUS REQ	55					
E11	ISTA WPBIT 0	57					
E12	ISTA WPBIT 1	61	0.0	-150.0	-60.0	mA	1-4
E13	ISTACK TO Y	63	0.0	-130.0	-30.0	mA	1-4
E14	P BIT 03	64	0.0	-150.0	-60.0	mA	1-4
E15	ISTA TO PIPE	65	0.0	-100.0	-40.0	mA	1-4
E16	P BIT 04	66	0.0	-150.0	-60.0	mA	1-4
E17	EWA PROTECT	69	0.0	-130.0	-30.0	mA	1-4
E18	AND CODE F	70	0.0	-130.0	-30.0	mA	1-4
E19	FLUSH ISTACK	71	0.0	-150.0	-60.0	mA	1-4
E20	IS RDPTR B1	79	0.0	-150.0	-60.0	mA	1-4
E21	SET ISTA EPT	82	0.0	-85.0	-15.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E22	JMP CON TRUE	88	0.0	-150.0	-60.0	mA	1-4
E23	JMP CON TRUE	88	0.0	-150.0	-60.0	mA	1-4
E24	CP MA CLR	93	0.0	-260.0	-30.0	mA	1-4
E25	ISTACK FULL	96	0.0	-150.0	-60.0	mA	1-4
E26	CB MIC CTRL	97	0.0	-150.0	-60.0	mA	1-4
E27	PROG RUN FF	98	0.0	-100.0	-40.0	mA	1-4
E28	ISTA RPBIT 0	99	0.0	-150.0	-60.0	mA	1-4

NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	E1-E28
Power supply current limit	1.93 A	NA	E1-E28
Forcing voltage, (Vout)	0.0 V	± 50 mV	E1-E28
Pattern voltage high, (VH)	3.0 V	± 100 mV	E1-E28
low, (VL)	0.0 V	± 100 mV	E1-E28

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output current measurement	NA	± 4.50 mA	E1
	NA	± 3.00 mA	E2
	NA	± 4.25 mA	E3
	NA	± 4.50 mA	E4,E5
	NA	± 4.25 mA	E6
	NA	± 5.00 mA	E7
	NA	± 4.25 mA	E8
	NA	± 4.50 mA	E9-E12
	NA	± 5.00 mA	E13
	NA	± 4.50 mA	E14
	NA	± 3.00 mA	E15
	NA	± 4.50 mA	E16
	NA	± 5.00 mA	E17,E18
	NA	± 4.50 mA	E19,E20
	NA	± 3.50 mA	E21
	NA	± 4.50 mA	E22,E23
	NA	± 11.50 mA	E24
	NA	± 4.50 mA	E25,E26
	NA	± 3.00 mA	E27
	NA	± 4.50 mA	E28

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

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NOTES: - Continued.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VI and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	MEM PROTECT	2	2.4	-1.2	0.04	mA	1-3
F2	CLII INTR EN	3	2.4	-1.1	0.07	mA	1-3
F3	CPMEM RES ER	4	2.4	-0.6	0.02	mA	1-3
F4	CB PARITY ER	5	2.4	-1.66	-0.54	mA	1-3
F5	DSCRETE MCLR	6	2.4	-3.06	-0.51	mA	1-3
F6	WRITE ISTACK	8	2.4	-0.8	0.04	mA	1-3
F7	SEL ISTK SEL	12	2.4	-0.6	0.02	mA	1-3
F8	BREAKPOINT	13	2.4	-1.0	0.04	mA	1-3
F9	MONITOR CLK	16	2.4	-1.31	-0.54	mA	1-3
F10	CIII INTR EN	17	2.4	-0.25	0.03	mA	1-3
F11	IR BIT 44	18	2.4	-1.7	0.09	mA	1-3
F12	MPIOMPC VINT	19	2.4	-0.25	0.03	mA	1-3
F13	IR BIT 06	20	2.4	-0.4	0.02	mA	1-3
F14	CK BRKPT STP	21	2.4	-2.6	0.07	mA	1-3
F15	CP CLOCK	26	2.4	-5.0	0.14	mA	1-3
F16	MIC CTRL FMT	27	2.4	-0.85	0.05	mA	1-3
F17	IR BIT 00	28	2.4	-0.6	0.02	mA	1-3
F18	IR BIT 01	29					
F19	IR BIT 02	30	2.4	-0.6	0.02	mA	1-3
F20	IR BIT 41	32	2.4	-0.2	0.02	mA	1-3
F21	IR BIT 08	33	2.4	-0.6	0.02	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F22	IR BIT 10	34	2.4	-0.6	0.02	mA	1-3
F23	IR BIT 13	35	2.4	-0.2	0.02	mA	1-3
F24	IR BIT 15	36	2.4	-0.8	0.04	mA	1-3
F25	MLTMATH PRES	37	2.4	-1.66	-0.54	mA	1-3
F26	SGR-EQ TO IR	38	2.4	-0.6	0.02	mA	1-3
F27	BOOT RTC PRS	39	2.4	-1.66	-0.54	mA	1-3
F28	CMI PRESENT	41	2.4	-1.66	-0.54	mA	1-3
F29	POWER FAULT	42	2.4	-1.26	-0.54	mA	1-3
F30	IOC CII INTR	43					
F31	MMIOCIIIINTR	44					
F32	DSCRETE STRT	45	2.4	-1.26	-0.54	mA	1-3
F33	STOIN ISTK D	46	2.4	-0.2	0.02	mA	1-3
F34	IOC COM BUSY	47	2.4	-1.26	-0.54	mA	1-3
F35	CPSC ABUS EN	48	2.4	-0.6	0.02	mA	1-3
F36	JMAP OE	49	2.4	-0.6	0.02	mA	1-3
F37	ECW 13	52	2.4	-0.85	0.05	mA	1-3
F38	ECW 12	53					
F39	ECW 15	54	2.4	-0.85	0.05	mA	1-3
F40	PWRUP MA CLR	56	2.4	-3.26	-0.49	mA	1-3
F41	IN ISTKW PTR	58	2.4	-1.68	0.08	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F42	CB CLK	59	2.4	-1.8	0.06	mA	1-3
F43	RTC OVFL INT	62	2.4	-1.31	-0.54	mA	1-3
F44	IR BIT 45	67	2.4	-0.85	0.05	mA	1-3
F45	CI INTR EN	68	2.4	-0.85	0.05	mA	1-3
F46	DEST BUS	72	2.4	-0.25	0.03	mA	1-3
F47	UIR 5	73	2.4	-0.5	0.05	mA	1-3
F48	KILL PIPE	74	2.4	-0.6	0.02	mA	1-3
F49	CKINS PI REG	76	2.4	-2.6	0.07	mA	1-3
F50	KIL PIPE	77	2.4	-2.05	0.09	mA	1-3
F51	FORC I FETCH	78	2.4	-0.85	0.05	mA	1-3
F52	IN ISTKR PTR	80	2.4	-2.28	0.1	mA	1-3
F53	IR BIT 07	83	2.4	-0.6	0.02	mA	1-3
F54	IR BIT 09	84	2.4	-0.6	0.02	mA	1-3
F55	IR BIT 12	85	2.4	-0.2	0.02	mA	1-3
F56	IR BIT 14	86	2.4	-0.8	0.04	mA	1-3
F57	DSCRT AUTOST	87	2.4	-1.66	-0.54	mA	1-3
F58	MAE PRESENT	89					
F59	IOC PRESENT	91	2.4	-1.66	-0.54	mA	1-3
F60	IOC CI INTR	92	2.4	-1.26	-0.54	mA	1-3
F61	IOC CIII INT	94					
F62	LOAD	95	2.4	-1.26	-0.54	mA	1-3

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NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F62
Power supply current limit	1.93 A	NA	F1-F62
Forcing voltage, (Vih)	2.4 V	± 50 mV	F1-F62
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 F1-F62 F1-F62
Input current measurement	NA	± 62.0 uA	F1
	NA	± 58.5 uA	F2
	NA	± 31.0 uA	F3
	NA	± 55.8 uA	F4
	NA	± 127.0 uA	F5
	NA	± 42.0 uA	F6
	NA	± 31.0 uA	F7
	NA	± 52.0 uA	F8
	NA	± 38.5 uA	F9
	NA	± 14.0 uA	F10
	NA	± 89.5 uA	F11
	NA	± 14.0 uA	F12
	NA	± 21.0 uA	F13
	NA	± 133.0 uA	F14
	NA	± 257.0 uA	F15
	NA	± 45.0 uA	F16
	NA	± 31.0 uA	F17-F19
	NA	± 11.0 uA	F20
	NA	± 31.0 uA	F21,F22
	NA	± 11.0 uA	F23
	NA	± 42.0 uA	F24
	NA	± 55.8 uA	F25

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input current measurement	NA	± 31.0 uA	F26
	NA	± 55.8 uA	F27,F28
	NA	± 35.8 uA	F29-F32
	NA	± 11.0 uA	F33
	NA	± 35.8 uA	F34
	NA	± 31.0 uA	F35,F36
	NA	± 45.0 uA	F37-F39
	NA	± 138.0 uA	F40
	NA	± 88.0 uA	F41
	NA	± 93.0 uA	F42
	NA	± 38.5 uA	F43
	NA	± 45.0 uA	F44,F45
	NA	± 13.7 uA	F46
	NA	± 27.5 uA	F47
	NA	± 31.0 uA	F48
	NA	± 133.0 uA	F49
	NA	± 107.0 uA	F50
	NA	± 45.0 uA	F51
	NA	± 119.0 uA	F52
	NA	± 31.0 uA	F53,F54
	NA	± 11.0 uA	F55
	NA	± 42.0 uA	F56
	NA	± 55.8 uA	F57-F59
	NA	± 35.8 uA	F60-F62

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

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NOTES: - Continued.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G1	MEM PROTECT	2	0.4	-1.2	0.04	mA	1-3
G2	CLII INTR EN	3	0.4	-1.1	0.07	mA	1-3
G3	CPMEM RES ER	4	0.4	-0.6	0.02	mA	1-3
G4	CB PARITY ER	5	0.4	-1.66	-0.54	mA	1-3
G5	DSCRETE MCLR	6	0.4	-3.06	-0.51	mA	1-3
G6	WRITE ISTACK	8	0.4	-0.8	0.04	mA	1-3
G7	SEL ISTK SEL	12	0.4	-0.6	0.02	mA	1-3
G8	BREAKPOINT	13	0.4	-1.0	0.04	mA	1-3
G9	MONITOR CLK	16	0.4	-1.31	-0.54	mA	1-3
G10	CIII INTR EN	17	0.4	-0.25	0.03	mA	1-3
G11	IR BIT 44	18	0.4	-1.7	0.09	mA	1-3
G12	MPIOMPC VINT	19	0.4	-0.25	0.03	mA	1-3
G13	IR BIT 06	20	0.4	-0.4	0.02	mA	1-3
G14	CK BRKPT STP	21	0.4	-2.6	0.07	mA	1-3
G15	CP CLOCK	26	0.4	-5.0	0.14	mA	1-3
G16	MIC CTRL FMT	27	0.4	-0.85	0.05	mA	1-3
G17	IR BIT 00	28	0.4	-0.6	0.02	mA	1-3
G18	IR BIT 01	29					
G19	IR BIT 02	30	0.4	-0.6	0.02	mA	1-3
G20	IR BIT 41	32	0.4	-0.2	0.02	mA	1-3
G21	IR BIT 08	33	0.4	-0.6	0.02	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G22	IR BIT 10	34	0.4	-0.6	0.02	mA	1-3
G23	IR BIT 13	35	0.4	-0.2	0.02	mA	1-3
G24	IR BIT 15	36	0.4	-0.8	0.04	mA	1-3
G25	MLTMATH PRES	37	0.4	-1.66	-0.54	mA	1-3
G26	SGR-EQ TO IR	38	0.4	-0.6	0.02	mA	1-3
G27	BOOT RTC PRS	39	0.4	-1.66	-0.54	mA	1-3
G28	CMI PRESENT	41	0.4	-1.66	-0.54	mA	1-3
G29	POWER FAULT	42	0.4	-1.26	-0.54	mA	1-3
G30	IOC CII INTR	43					
G31	MMIOCIIIINTR	44					
G32	DSCRETE STRT	45	0.4	-1.26	-0.54	mA	1-3
G33	STOIN ISTK D	46	0.4	-0.2	0.02	mA	1-3
G34	IOC COM BUSY	47	0.4	-1.26	-0.54	mA	1-3
G35	CPSC ABUS EN	48	0.4	-0.6	0.02	mA	1-3
G36	JMAP OE	49	0.4	-0.6	0.02	mA	1-3
G37	ECW 13	52	0.4	-0.85	0.05	mA	1-3
G38	ECW 12	53					
G39	ECW 15	54	0.4	-0.85	0.05	mA	1-3
G40	PWRUP MA CLR	56	0.4	-3.26	-0.49	mA	1-3
G41	IN ISTKW PTR	58	0.4	-1.68	0.08	mA	1-3
G42	CB CLK	59	0.4	-1.8	0.06	mA	1-3

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Check the source to verify that this is the current version before use.

TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G43	RTC OVFL INT	62	0.4	-1.31	-0.54	mA	1-3
G44	IR BIT 45	67	0.4	-0.85	0.05	mA	1-3
G45	CI INTR EN	68	0.4	-0.85	0.05	mA	1-3
G46	DEST BUS	72	0.4	-0.25	0.03	mA	1-3
G47	UIR 5	73	0.4	-0.5	0.05	mA	1-3
G48	KILL PIPE	74	0.4	-0.6	0.02	mA	1-3
G49	CKINS PI REG	76	0.4	-2.6	0.07	mA	1-3
G50	KIL PIPE	77	0.4	-2.05	0.09	mA	1-3
G51	FORC I FETCH	78	0.4	-0.85	0.05	mA	1-3
G52	IN ISTKR PTR	80	0.4	-2.28	0.1	mA	1-3
G53	IR BIT 07	83	0.4	-0.6	0.02	mA	1-3
G54	IR BIT 09	84	0.4	-0.6	0.02	mA	1-3
G55	IR BIT 12	85	0.4	-0.2	0.02	mA	1-3
G56	IR BIT 14	86	0.4	-0.8	0.04	mA	1-3
G57	DSCRT AUTOST	87	0.4	-1.66	-0.54	mA	1-3
G58	MAE PRESENT	89					
G59	IOC PRESENT	91	0.4	-1.66	-0.54	mA	1-3
G60	IOC CI INTR	92	0.4	-1.26	-0.54	mA	1-3
G61	IOC CIII INT	94					
G62	LOAD	95	0.4	-1.26	-0.54	mA	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:44Z
Check the source to verify that this is the current version before use.

NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G62
Power supply current limit	1.93 A	NA	G1-G62
Forcing voltage, (Vil)	0.4 V	± 50 mV	G1-G62
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	G1-G62
low, (VL)	0.0 V	± 100 mV	G1-G62
Input current measurement	NA	± 62.0 uA	G1
	NA	± 58.5 uA	G2
	NA	± 31.0 uA	G3
	NA	± 55.8 uA	G4
	NA	± 127.0 uA	G5
	NA	± 42.0 uA	G6
	NA	± 31.0 uA	G7
	NA	± 52.0 uA	G8
	NA	± 38.5 uA	G9
	NA	± 14.0 uA	G10
	NA	± 89.5 uA	G11
	NA	± 14.0 uA	G12
	NA	± 21.0 uA	G13
	NA	± 133.0 uA	G14
	NA	± 257.0 uA	G15
	NA	± 45.0 uA	G16
	NA	± 31.0 uA	G17-G19
	NA	± 11.0 uA	G20
	NA	± 31.0 uA	G21,G22
	NA	± 11.0 uA	G23
	NA	± 42.0 uA	G24
	NA	± 55.8 uA	G25

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:44Z
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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input current measurement	NA	± 31.0 uA	G26
	NA	± 55.8 uA	G27,G28
	NA	± 35.8 uA	G29-G32
	NA	± 11.0 uA	G33
	NA	± 35.8 uA	G34
	NA	± 31.0 uA	G35,G36
	NA	± 45.0 uA	G37-G39
	NA	± 138.0 uA	G40
	NA	± 88.0 uA	G41
	NA	± 93.0 uA	G42
	NA	± 38.5 uA	G43
	NA	± 45.0 uA	G44,G45
	NA	± 13.7 uA	G46
	NA	± 27.5 uA	G47
	NA	± 31.0 uA	G48
	NA	± 133.0 uA	G49
	NA	± 107.0 uA	G50
	NA	± 45.0 uA	G51
	NA	± 119.0 uA	G52
	NA	± 31.0 uA	G53,G54
	NA	± 11.0 uA	G55
	NA	± 42.0 uA	G56
	NA	± 55.8 uA	G57-G59
	NA	± 35.8 uA	G60-G62

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H1	CB MA CLR	7	-0.4	2.4	4.5	V	1-4
H2	TP	9	-0.4	2.4	4.5	V	1-4
H3	NDRO BIT MOD	11	-0.16	2.4	4.5	V	1-4
H4	P BIT 06	14	-0.4	2.4	4.5	V	1-4
H5	P BIT 05	15	-0.4	2.4	4.5	V	1-4
H6	PRG FAULT FF	22	-0.16	2.4	4.5	V	1-4
H7	INTR PENDING	23					
H8	RST2 RNK FUL	24	-0.16	2.4	4.5	V	1-4
H9	IN IOC UPD	50	-0.4	2.4	4.5	V	1-4
H10	SET BUS REQ	55					
H11	ISTA WPBIT 0	57					
H12	ISTA WPBIT 1	61	-0.4	2.4	4.5	V	1-4
H13	ISTACK TO Y	63	-0.16	2.4	4.5	V	1-4
H14	P BIT 03	64	-0.4	2.4	4.5	V	1-4
H15	ISTA TO PIPE	65					
H16	P BIT 04	66	-0.4	2.4	4.5	V	1-4
H17	EWA PROTECT	69	-0.16	2.4	4.5	V	1-4
H18	AND CODE F	70	-0.16	2.4	4.5	V	1-4
H19	FLUSH ISTACK	71	-0.4	2.4	4.5	V	1-4
H20	IS RDPTR B1	79	-0.4	2.4	4.5	V	1-4
H21	SET ISTA EPT	82	-0.16	2.4	4.5	V	1-4

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Check the source to verify that this is the current version before use.

TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H22	JMP CON TRUE	88	-0.4	2.4	4.5	V	1-4
H23	JMP CON TRUE	88	-0.4	2.4	4.5	V	1-4
H24	CP MA CLR	93	-0.16	2.4	4.5	V	1-4
H25	ISTACK FULL	96	-0.4	2.4	4.5	V	1-4
H26	CB MIC CTRL	97					
H27	PROG RUN FF	98					
H28	ISTA RPBIT 0	99	-0.4	2.4	4.5	V	1-4

NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	H1-H28
Power supply current limit	1.93 A	NA	H1-H28
Forcing current, (Ioh)	-0.4 mA	± 40 uA	H1,H2
	-0.16 mA	± 16 uA	H3
	-0.4 mA	± 40 uA	H4,H5
	-0.16 mA	± 16 uA	H6-H8
	-0.4 mA	± 40 uA	H9-H12
	-0.16 mA	± 16 uA	H13
	-0.4 mA	± 40 uA	H14-H16
	-0.16 mA	± 16 uA	H17,H18
	-0.4 mA	± 40 uA	H19,H20
	-0.16 mA	± 16 uA	H21

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Check the source to verify that this is the current version before use.

NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Forcing current, (Ioh)	-0.4 mA -0.16 mA -0.4 mA	± 40 uA ± 16 uA ± 40 uA	H22,H23 H24 H25-H28
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	H1-H28 H1-H28
Output voltage measurement	NA	± 105 mV	H1-H28

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

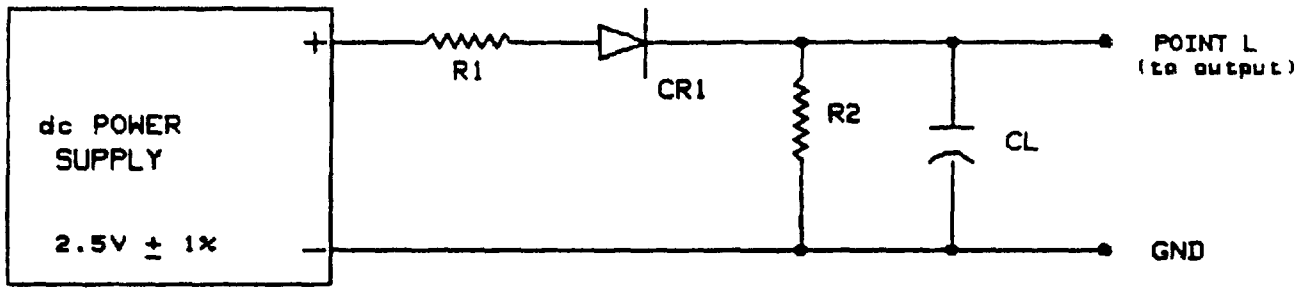
(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table IX and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

    (a) C = 100 ± 10 pF (including probe and parasitic capacitance) for sample dynamic tests.

    (b) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

    (c) The functional test shall be performed with only a 100 ± 10 pF load (including probe and parasitic capacitance) attached to each output.

    (d) C not required for any other tests.

FIGURE 9. Logic load circuit.

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NOTES: - Continued.

3. Select R1 for a current flow of Iol when point L is held at Vol. The values of Iol and Vol for each output are in the table below.

4. The value of R2 for each output load shall be as specified in the table below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
7,9,14,15,50,55, 57,61,64,65,66, 71,79,88,96,97, 98,99	6200	6.0	0.5
11,22,23,24,63, 69,70,82,93	16000	1.6	0.5

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE X. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J1	CB MA CLR	7	6.0	0.0	0.5	V	1-4
J2	TP	9	6.0	0.0	0.5	V	1-4
J3	NDRO BIT MOD	11	1.6	0.0	0.5	V	1-4
J4	P BIT 06	14	6.0	0.0	0.5	V	1-4
J5	P BIT 05	15	6.0	0.0	0.5	V	1-4
J6	PRG FAULT FF	22	1.6	0.0	0.5	V	1-4
J7	INTR PENDING	23					
J8	RST2 RNK FUL	24	1.6	0.0	0.5	V	1-4
J9	IN IOC UPD	50	6.0	0.0	0.5	V	1-4
J10	SET BUS REQ	55					
J11	ISTA WPBIT 0	57					
J12	ISTA WPBIT 1	61	6.0	0.0	0.5	V	1-4
J13	ISTACK TO Y	63	1.6	0.0	0.5	V	1-4
J14	P BIT 03	64	6.0	0.0	0.5	V	1-4
J15	ISTA TO PIPE	65					
J16	P BIT 04	66	6.0	0.0	0.5	V	1-4
J17	EWA PROTECT	69	1.6	0.0	0.5	V	1-4
J18	AND CODE F	70	1.6	0.0	0.5	V	1-4
J19	FLUSH ISTACK	71	6.0	0.0	0.5	V	1-4
J20	IS RDPTR B1	79	6.0	0.0	0.5	V	1-4
J21	SET ISTA EPT	82	1.6	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J22	JMP CON TRUE	88	6.0	0.0	0.5	V	1-4
J23	JMP CON TRUE	88	6.0	0.0	0.5	V	1-4
J24	CP MA CLR	93	1.6	0.0	0.5	V	1-4
J25	ISTACK FULL	96	6.0	0.0	0.5	V	1-4
J26	CB MIC CTRL	97					
J27	PROG RUN FF	98					
J28	ISTA RPBIT 0	99	6.0	0.0	0.5	V	1-4

NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J28
Power supply current limit	1.93 A	NA	J1-J28
Forcing current, (Iol)	6.0 mA	± 600 uA	J1,J2
	1.6 mA	± 160 uA	J3
	6.0 mA	± 600 uA	J4,J5
	1.6 mA	± 160 uA	J6-J8
	6.0 mA	± 600 uA	J9-J12
	1.6 mA	± 160 uA	J13
	6.0 mA	± 600 uA	J14-J16
	1.6 mA	± 160 uA	J17,J18
	6.0 mA	± 600 uA	J19,J20
	1.6 mA	± 160 uA	J21

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Forcing current, (Iol)	6.0 mA 1.6 mA 6.0 mA	± 600 uA ± 160 uA ± 600 uA	J22,J23 J24 J25-J28
Pattern voltage, high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	J1-J28 J1-J28
Output voltage measurement	NA	± 25 mV	J1-J28

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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TABLE XI. Output leakage current, high and low level, test.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
High level							
K1	P BIT 06	14	2.4	-0.05	0.05	mA	1-3
K2	P BIT 05	15					
K3	P BIT 03	64					
K4	P BIT 04	66					
K5	JMP CON TRUE	88	2.4	-0.05	0.05	mA	1-3
Low level							
K6	P BIT 06	14	0.4	-0.05	0.05	mA	1-3
K7	P BIT 05	15					
K8	P BIT 03	64					
K9	P BIT 04	66					
K10	JMP CON TRUE	88	0.4	-0.05	0.05	mA	1-3

NOTES:

1. Output leakage current, high and low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	K1-K10
Power supply current limit	1.93 A	NA	K1-K10

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Forcing voltage, (Voh) (Vol)	 2.4 V 0.4 V	 ± 50 mV ± 50 mV	 K1-K5 K6-K10
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 K1-K10 K1-K10
Output current measurement	NA	± 5 uA	K1-K10

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XI and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XII. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L1	MEM PROTECT	2	-18.0	-1.2	0.0	V	1-3
L2	CLII INTR EN	3					
L3	CPMEM RES ER	4	-18.0	-1.2	0.0	V	1-3
L4	WRITE ISTACK	8	-18.0	-1.5	0.0	V	1-3
L5	SEL ISTK SEL	12	-18.0	-1.2	0.0	V	1-3
L6	BREAKPOINT	13	-18.0	-1.2	0.0	V	1-3
L7	CIII INTR EN	17	-18.0	-1.5	0.0	V	1-3
L8	IR BIT 44	18					
L9	MPIOMPC VINT	19					
L10	IR BIT 06	20	-18.0	-1.5	0.0	V	1-3
L11	CK BRKPT STP	21	-18.0	-1.2	0.0	V	1-3
L12	CP CLOCK	26					
L13	MIC CTRL FMT	27					
L14	IR BIT 00	28					
L15	IR BIT 01	29					
L16	IR BIT 02	30	-18.0	-1.2	0.0	V	1-3
L17	IR BIT 41	32	-18.0	-1.5	0.0	V	1-3
L18	IR BIT 08	33	-18.0	-1.2	0.0	V	1-3
L19	IR BIT 10	34	-18.0	-1.2	0.0	V	1-3
L20	IR BIT 13	35	-18.0	-1.5	0.0	V	1-3
L21	IR BIT 15	36	-18.0	-1.2	0.0	V	1-3

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TABLE XII. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L22	SGR-EQ TO IR	38	-18.0	-1.2	0.0	V	1-3
L23	STOIN ISTK D	46	-18.0	-1.5	0.0	V	1-3
L24	CPSC ABUS EN	48	-18.0	-1.2	0.0	V	1-3
L25	JMAP OE	49	-18.0	-1.2	0.0	V	1-3
L26	ECW 13	52	-18.0	-1.5	0.0	V	1-3
L27	ECW 12	53					
L28	ECW 15	54	-18.0	-1.5	0.0	V	1-3
L29	IN ISTKW PTR	58	-18.0	-1.2	0.0	V	1-3
L30	CB CLK	59	-18.0	-1.2	0.0	V	1-3
L31	IR BIT 45	67	-18.0	-1.5	0.0	V	1-3
L32	CI INTR EN	68					
L33	DEST BUS	72					
L34	UIR 5	73	-18.0	-1.5	0.0	V	1-3
L35	KILL PIPE	74	-18.0	-1.2	0.0	V	1-3
L36	CKINS PI REG	76					
L37	KIL PIPE	77					
L38	FORC I FETCH	78	-18.0	-1.2	0.0	V	1-3
L39	IN ISTKR PTR	80	-18.0	-1.5	0.0	V	1-3
L40	IR BIT 07	83	-18.0	-1.2	0.0	V	1-3

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TABLE XII. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L41	IR BIT 09	84	-18.0	-1.2	0.0	V	1-3
L42	IR BIT 12	85	-18.0	-1.5	0.0	V	1-3
L43	IR BIT 14	86	-18.0	-1.5	0.0	V	1-3

NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	Off	NA	L1-L43
Power supply current limit	NA	NA	L1-L43
Forcing current, (Iin)	-18 mA	± 1.8 mA	L1-L43
Pattern voltage high, (VH)	NA	NA	L1-L43
low, (VL)	NA	NA	L1-L43
Input voltage measurement	NA	± 60 mV	L1-L3
	NA	± 75 mV	L4
	NA	± 60 mV	L5,L6
	NA	± 75 mV	L7-L10
	NA	± 60 mV	L11-L16
	NA	± 75 mV	L17
	NA	± 60 mV	L18,L19
	NA	± 75 mV	L20

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input voltage measurement	NA	± 60 mV	L21,L22
	NA	± 75 mV	L23
	NA	± 60 mV	L24,L25
	NA	± 75 mV	L26-L28
	NA	± 60 mV	L29,L30
	NA	± 75 mV	L31-L34
	NA	± 60 mV	L35-L38
	NA	± 75 mV	L39
	NA	± 60 mV	L40,L41
	NA	± 75 mV	L42,L43

2. The following procedure is required to perform this test:

(a) No power shall be applied to the module.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XII and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE XIII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
M1	PSFC	1	5.1	9.5	uF	1-4
M2	PSFC	51	7.8	14.6	uF	1-4

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	M1,M2
Capacitance bridge voltage	0.3 Vp	± 30 mVp	M1,M2
Capacitance measurement	NA NA	± 220 nF ± 340 nF	M1 M2

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XIII.

3. The test circuit shall be as indicated on figure 10.

4. Abbreviations used are defined in 6.1.

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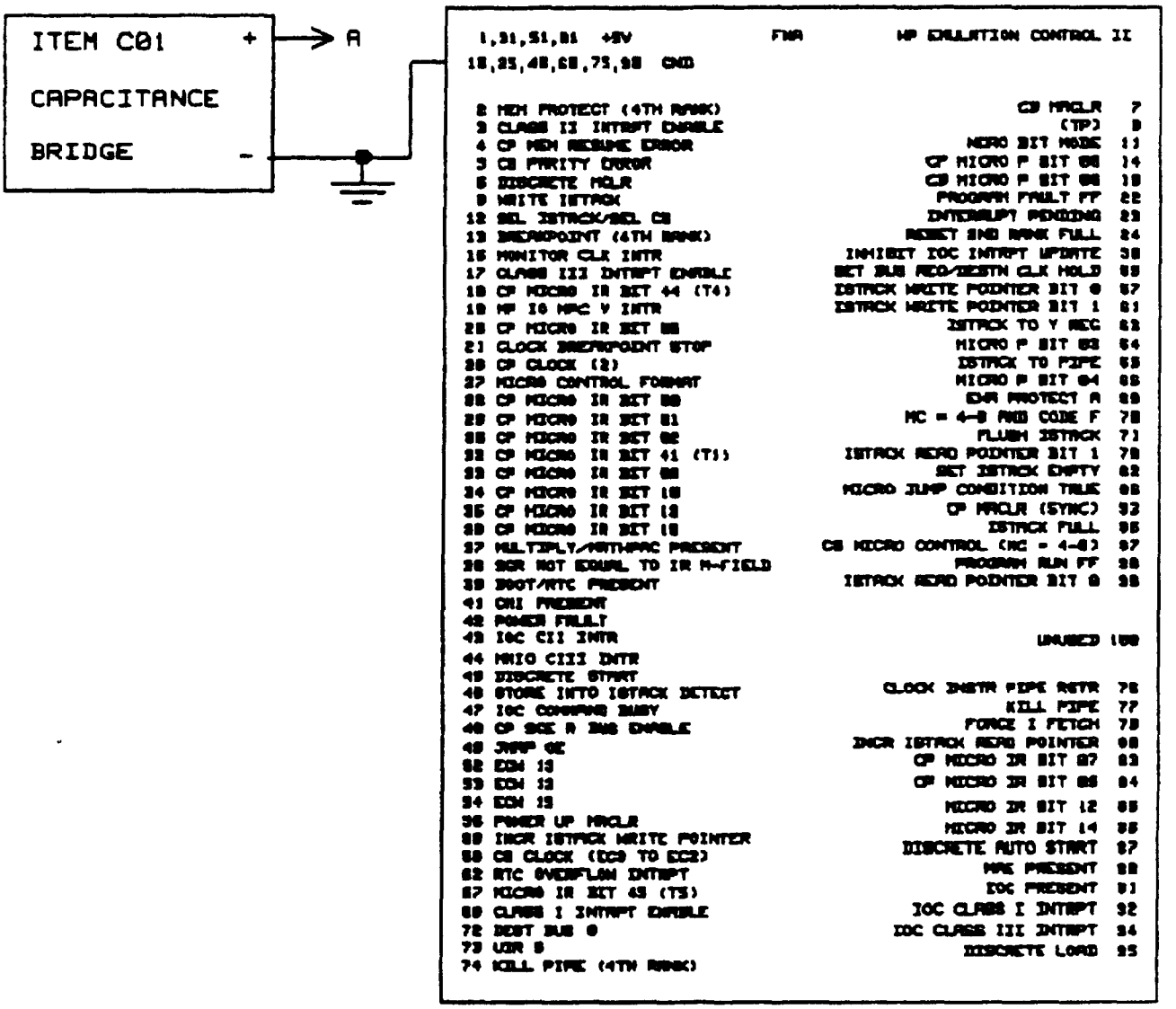


FIGURE 10. Power supply filter capacitance test circuit for table XIII.

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TABLE XIV. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
N	Icc	1,31, 51,81	856	1284	mA	1-3

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N
Power supply current limit	1.93 A	NA	N
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 N N
Power supply current measurement	NA	± 21.4 mA	N

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIV.

3. Abbreviations used are defined in 6.1.

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TABLE XV. Functional test.

Test cond	Test description	Notes	Figs
P1	Functional test minimum power supply voltage maximum pattern rate (225 nS)	1-3	9,11
P2	Functional test maximum power supply voltage maximum pattern rate (225 nS)	1-3	9,11

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	P1 P2
Power supply current limit	1.93 A	NA	P1,P2
Pattern voltage, high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 P1,P2 P1,P2
Output voltage verification, high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 105 mV ± 155 mV ± 25 mV	 P1 P2 P1,P2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XV using only a capacitive load of 100 pF (see figure 9). The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.5\ V$

(c) The module inputs shall not violate the minimum input timing parameters referenced on figure 11.

3. Abbreviations used are defined in 6.1.

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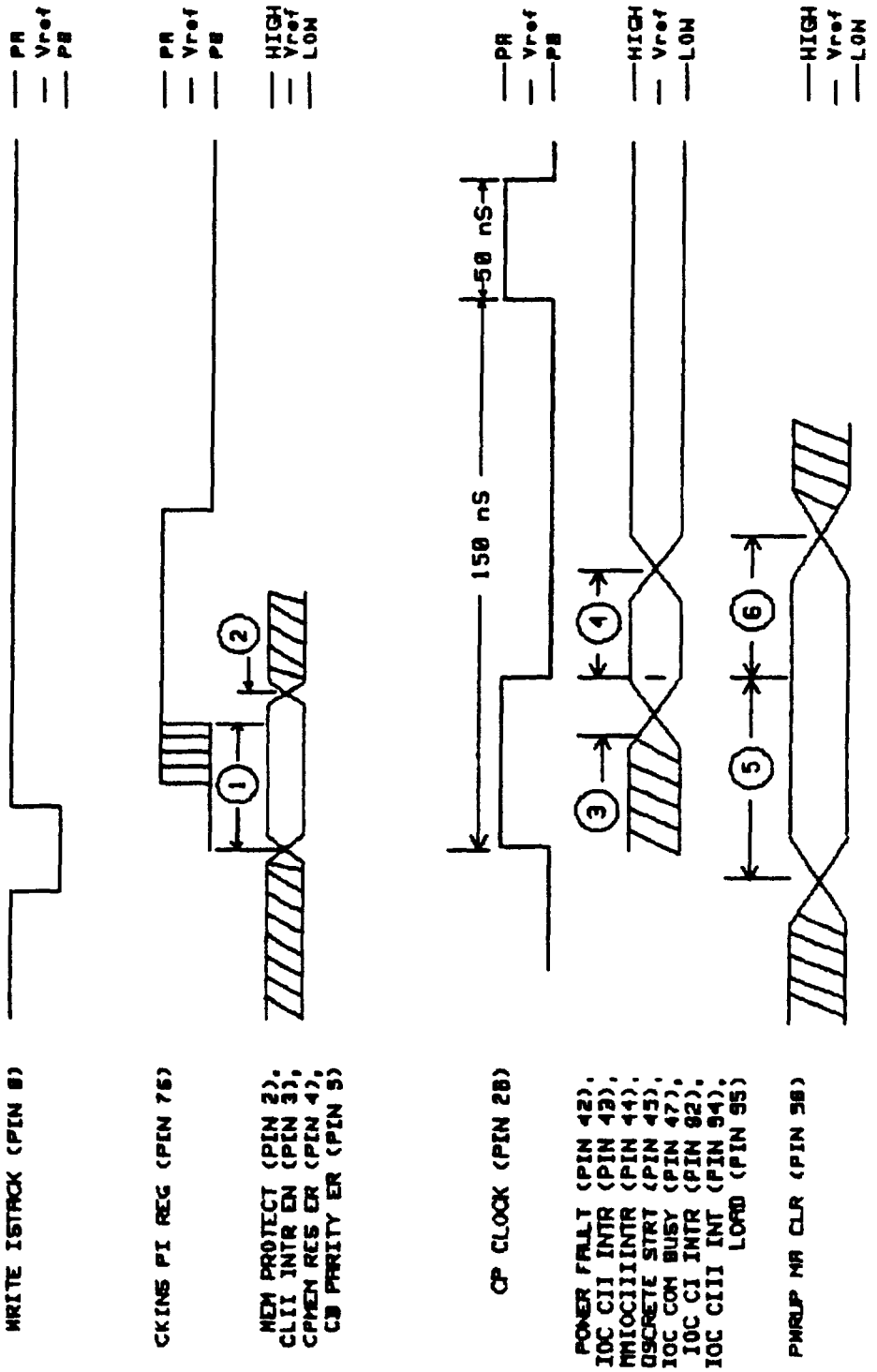


FIGURE 11a. Input timing parameters for table XV.

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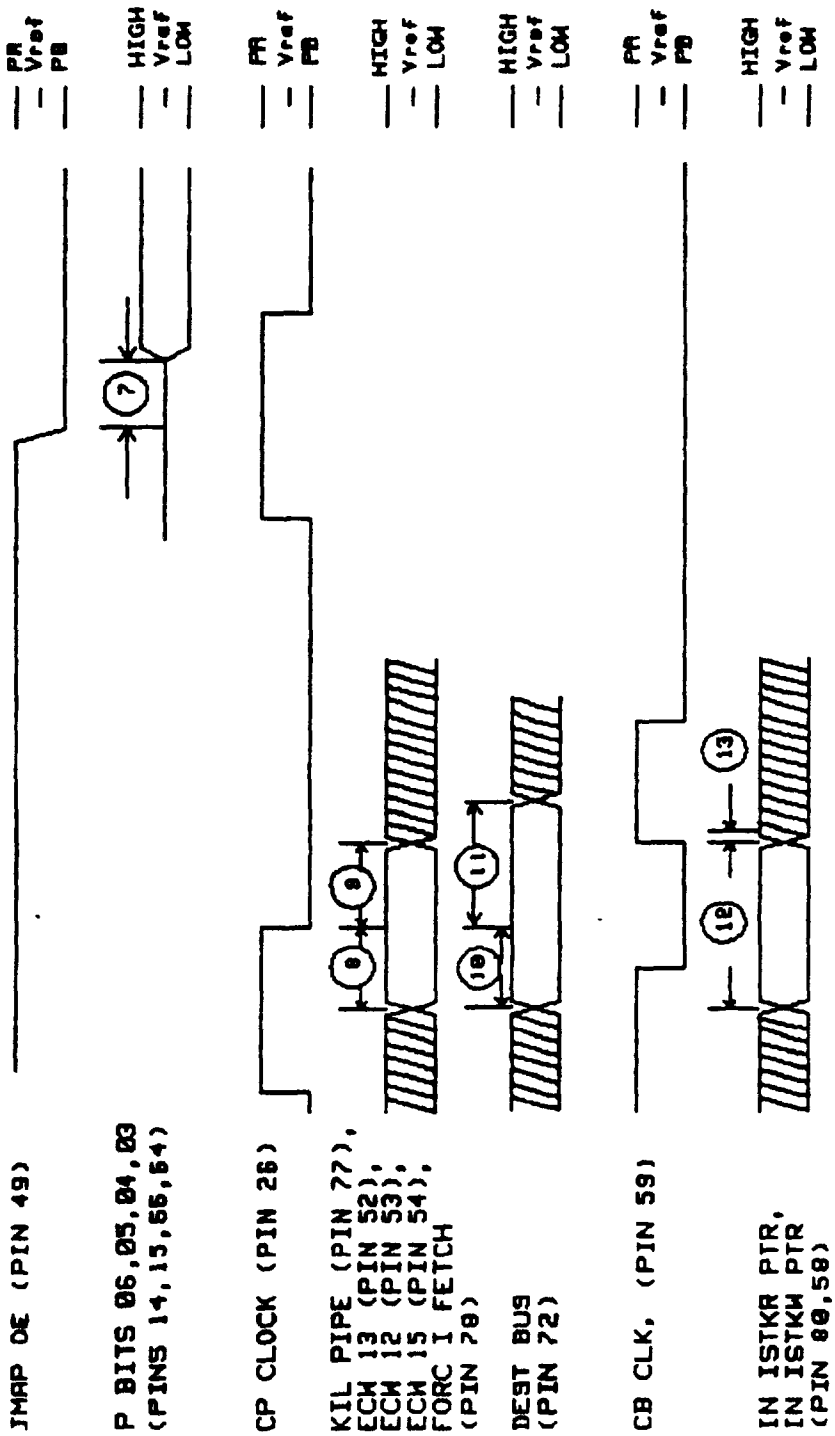


FIGURE 11b. Input timing parameters for table XV.

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NOTES:

1. (1) MEM PROTECT, CLII INTR EN, CPMEM RES ER, CB PARITY ER to CKINS PI REG. Setup time: 37 nS minimum.

2. (2) CKINS PI REG to MEM PROTECT, CLII INTR EN, CPMEM RES ER, CB PARITY ER. Hold time: 7 nS minimum.

3. (3) POWER FAULT, IOC CII INTR, MMIOCIIIINTR, DSCRETE STRT, IOC COM BUSY, IOC CI INTR, IOC CIII INT, LOAD to CP CLOCK. Setup time: 7 nS minimum.

4. (4) CP CLOCK to POWER FAULT, IOC CII INTR, MMIOCIIIINT, DSCRETE STRT, IOC COM BUSY, IOC CI INTR, IOC CIII INTR, LOAD. Hold time: 14 nS minimum.

5. (5) PWRUP MA CLR to CP CLOCK. Setup time: 29 nS minimum.

6. (6) CP CLOCK to PWRUP MA CLR. Hold time: 9 nS minimum.

7. (7) JMAP OE to P BIT 06, P BIT 05, P BIT 03, P BIT 04. Tristate enable time: 27 nS maximum.

8. (8) ECW 13, ECW 12, ECW 15, KIL PIPE to CP CLOCK. Setup time: 65 nS minimum.

9. (9) CP CLOCK to ECW 13, ECW 12, ECW 15, KIL PIPE, FORC I FETCH. Hold time: 10 nS minimum.

10. (10) DEST BUS to CP CLOCK. Setup time: 42 nS minimum.

11. (11) CP CLOCK to DEST BUS. Hold time: 10 nS minimum.

12. (12) IN ISTKR PTR, IN ISTKW PTR to CB CLOCK. Setup time: 24 nS minimum.

13. (13) CB CLOCK to IN ISTKR PTR, IN ISTKW PTR. Hold time: 1 nS minimum.

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TABLE XVI. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate (225 nS) minimum power supply	1-6
Q2	Dynamic test, sample maximum pattern rate (225 nS) maximum power supply	1-6
Q3	Dynamic test, qualification maximum pattern rate (225 nS) minimum power supply	1-6
Q4	Dynamic test, qualification maximum pattern rate (225 nS) maximum power supply	1-6

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Current limit	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	1.93 A 1.93 A	Q1,Q3 Q2,Q4
Input voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 NA NA	 Q1-Q4 Q1-Q4

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NOTES: - Continued.

Condition	Value	Accuracy	Current limit	Test condition
Output voltage verification				
high, (Voh)	NA	± 100 mV	NA	Q2,Q4
high, (Voh)	NA	± 100 mV	NA	Q1,Q3
low, (Vol)	NA	± 100 mV	NA	Q1-Q4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

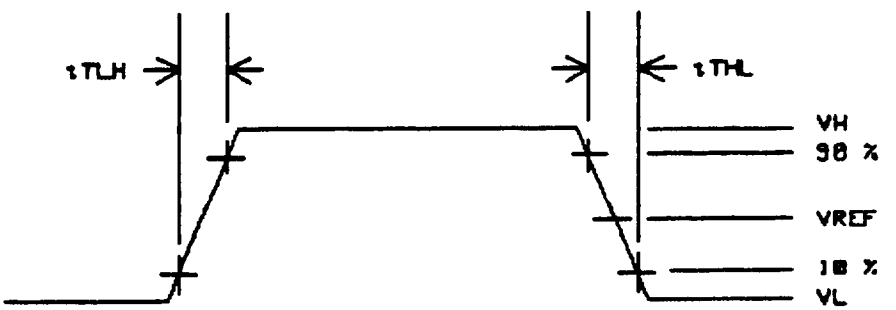
(b) The output compare levels shall be as follows:

High level: $Vref \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq Vref$

(c) The dynamic parameters are found in table XVII.

3. The load circuit for this test is specified on figure 9.

4. Input/Output (I/O) waveforms are as referenced on figure 12. All of the input waveforms shall have the following characteristics:



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NOTES: - Continued.

Where:

(a) VH and VL are as specified in the test tolerance table.

(b) tTLH = tTHL $\leq$ 10 nS.

(c) Vref = 1.6 V.

5. This test shall be performed as indicated in table XVI.

6. Abbreviations used are defined in 6.1.

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TABLE XVII. Dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	CP CLK to NDRO BIT MOD	-	-	-	72	72	72	nS	12
1	tpd	CP CLK to PROG RUN FF	-	-	-	72	72	72	nS	12
2	tpd	CP CLK to RST2 RNK FUL	-	-	-	52	52	52	nS	12
3	tpd	CP CLK to IN IOC UPD	-	-	-	25	25	25	nS	12
4	tpd	SGR-EQ TO IR to JMP CON TRUE	-	-	-	16	16	16	nS	12
5	tpd	CP CLK to CP MA CLR	-	-	-	29	29	29	nS	12
6	tpd	PWRUP MA CLR to CB MA CLR	-	-	-	22	22	22	nS	12
6	tpd	DSCRETE MCLR to CB MA CLR	-	-	-	22	22	22	nS	12
7	tpd	IR BIT 44 to INTR PENDING	-	-	-	90	90	90	nS	12
8	tpd	CK BRKPT STP to INTR PENDING	-	-	-	71	71	71	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
9	tpd	IR BIT 44 to P BIT 06	-	-	-	101	101	101	nS	12
9	tpd	IR BIT 44 to P BIT 05	-	-	-	101	101	101	nS	12
9	tpd	IR BIT 44 to P BIT 03	-	-	-	101	101	101	nS	12
9	tpd	IR BIT 44 to P BIT 04	-	-	-	101	101	101	nS	12
10	tpd	CP CLK to SET BUS REQ	-	-	-	92	92	92	nS	12
11	tpd	MIC CTRL FMT to CB MIC CTRL	-	-	-	36	36	36	nS	12
12	tpd	IR BIT 09 to EWA PROTECT	-	-	-	58	58	58	nS	12
13	tpd	CP CLK to ISTACK TO Y	-	-	-	92	92	92	nS	12
14	tpd	KILL PIPE to FLUSH ISTACK	-	-	-	13	13	13	nS	12
15	tpd	CP CLK to FLUSH ISTACK	-	-	-	26	26	26	nS	12
16	tpd	CB CLK to ISTACK FULL	-	-	-	15	15	15	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
17	tpd	CB CLK to ISTACK WPBIT 1	-	-	-	13	13	13	nS	12
17	tpd	CB CLK to ISTACK WPBIT 0	-	-	-	13	13	13	nS	12
17	tpd	CB CLK to IS RDPTR B1	-	-	-	13	13	13	nS	12
17	tpd	CB CLK to ISTA RPBIT 0	-	-	-	13	13	13	nS	12
18	tpd	CB CLK to SET ISTA EPT	-	-	-	45	45	45	nS	12
19	tpd	IN ISTKR PTR to SET ISTA EPT	-	-	-	22	22	22	nS	12
19	tpd	IN ISTKW PTR to SET ISTA EPT	-	-	-	22	22	22	nS	12
20	tsu	MEM PRO-TECT to CKINS PI REG	37	37	37	-	-	-	nS	12
20	tsu	CLII INTR EN to CKINS PI REG	37	37	37	-	-	-	nS	12
20	tsu	CPMEM RES ER to CKINS PI REG	37	37	37	-	-	-	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
20	tsu	CB PARITY ER to CKINS PI REG	37	37	37	-	-	-	nS	12
21	th	CKINS PI REG to MEM PRO-TECT	7	7	7	-	-	-	nS	12
21	th	CKINS PI REG to CLII INTR EN	7	7	7	-	-	-	nS	12
21	th	CKINS PI REG to CPMEM RES ER	7	7	7	-	-	-	nS	12
21	th	CKINS PI REG to CB PARITY ER	7	7	7	-	-	-	nS	12
22	tpd	CPSC ABUS EN to JMP CON TRUE	-	-	-	28	28	28	nS	12
23	tsu	POWER FAULT to CP CLOCK	7	7	7	-	-	-	nS	12
23	tsu	IOC CII INTR to CP CLOCK	7	7	7	-	-	-	nS	12
23	tsu	MMIOCIII-INTR to CP CLOCK	7	7	7	-	-	-	nS	12
23	tsu	DSCRETE STRT to CP CLOCK	7	7	7	-	-	-	nS	12
23	tsu	IOC COM BUSY to CP CLOCK	7	7	7	-	-	-	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
23	tsu	IOC CI INTR to CP CLOCK	7	7	7	-	-	-	nS	12
23	tsu	IOC CIII INT to CP CLOCK	7	7	7	-	-	-	nS	12
23	tsu	LOAD to CP CLOCK	7	7	7	-	-	-	nS	12
24	th	CP CLOCK to POWER FAULT	14	14	14	-	-	-	nS	12
24	th	CP CLOCK to IOC CII INTR	14	14	14	-	-	-	nS	12
24	th	CP CLOCK to MMIOC-IIIINTR	14	14	14	-	-	-	nS	12
24	th	CP CLOCK to DSCRETE STRT	14	14	14	-	-	-	nS	12
24	th	CP CLOCK to IOC COM BUSY	14	14	14	-	-	-	nS	12
24	th	CP CLOCK to IOC CI INTR	14	14	14	-	-	-	nS	12
24	th	CP CLOCK to IOC CIII INTR	14	14	14	-	-	-	nS	12
24	th	CP CLOCK to LOAD	14	14	14	-	-	-	nS	12
25	tsu	PWRUP MA CLR to CP CLOCK	29	29	29	-	-	-	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
26	th	CP CLOCK to PWRUP MA CLR	9	9	9	-	-	-	nS	12
27	te	JMAP OE to P BIT 06	-	-	-	27	27	27	nS	12
27	te	JMAP OE to P BIT 05	-	-	-	27	27	27	nS	12
27	te	JMAP OE to P BIT 03	-	-	-	27	27	27	nS	12
27	te	JMAP OE to P BIT 04	-	-	-	27	27	27	nS	12
28	tsu	ECW 13 to CP CLOCK	65	65	65	-	-	-	nS	12
28	tsu	ECW 12 to CP CLOCK	65	65	65	-	-	-	nS	12
28	tsu	ECW 15 to CP CLOCK	65	65	65	-	-	-	nS	12
28	tsu	KIL PIPE to CP CLOCK	65	65	65	-	-	-	nS	12
29	th	CP CLOCK to ECW 13	10	10	10	-	-	-	nS	12
29	th	CP CLOCK to ECW 12	10	10	10	-	-	-	nS	12
29	th	CP CLOCK to ECW 15	10	10	10	-	-	-	nS	12
29	th	CP CLOCK to KIL PIPE	10	10	10	-	-	-	nS	12
29	th	CP CLOCK to FORC I FETCH	10	10	10	-	-	-	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref Num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
30	tsu	DEST BUS to CP CLOCK	42	42	42	-	-	-	nS	12
31	th	CP CLK to DEST BUS	10	10	10	-	-	-	nS	12
32	tsu	IN ISTKR PTR to CB CLOCK	24	24	24	-	-	-	nS	12
32	tsu	IN ISTKW PTR to CB CLOCK	24	24	24	-	-	-	nS	12
33	th	CB CLOCK to IN ISTKR PTR	1	1	1	-	-	-	nS	12
	th	CB CLOCK to IN ISTKW PTR	1	1	1	-	-	-	nS	12

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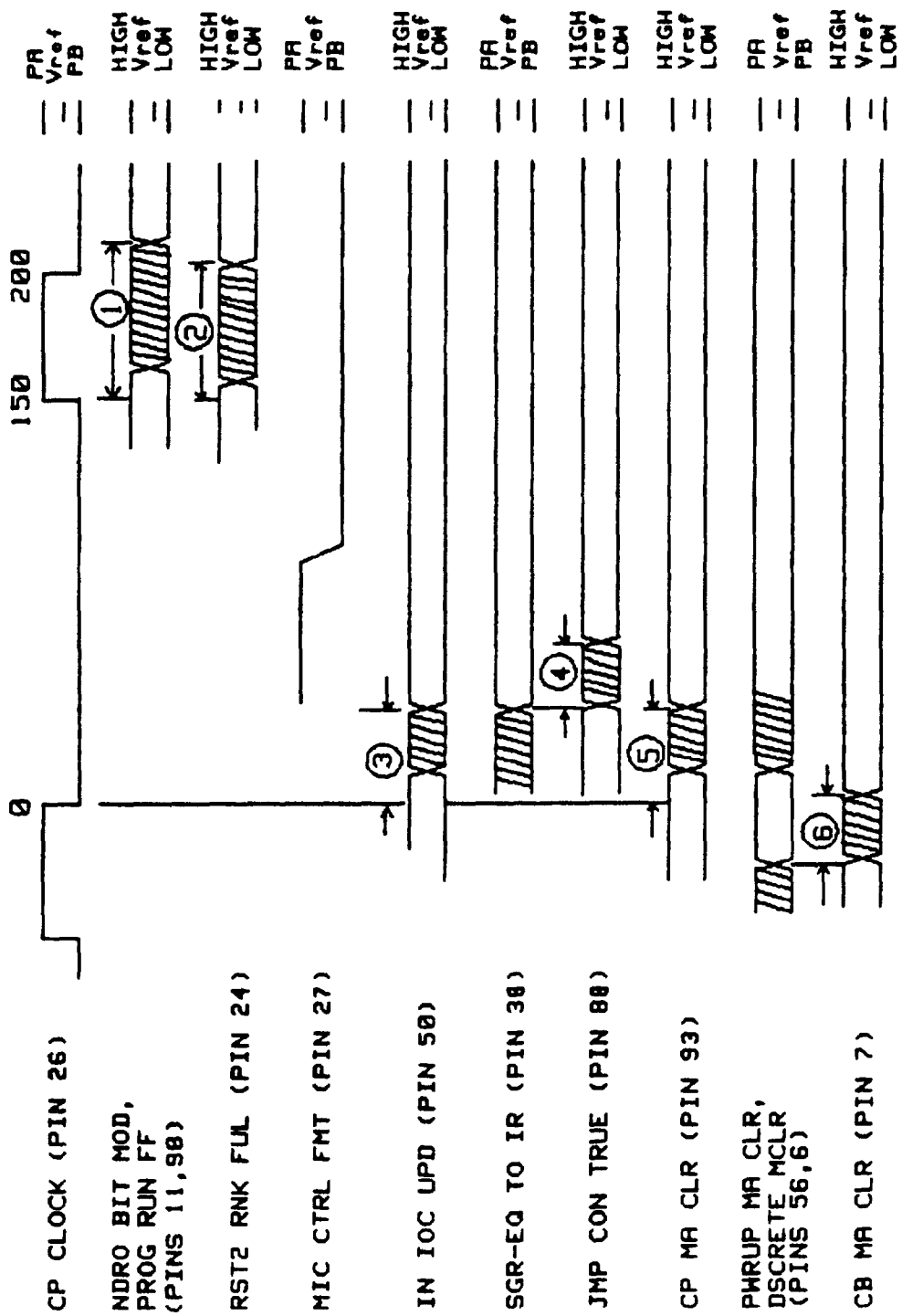


FIGURE 12a. Dynamic waveforms for table XVII.

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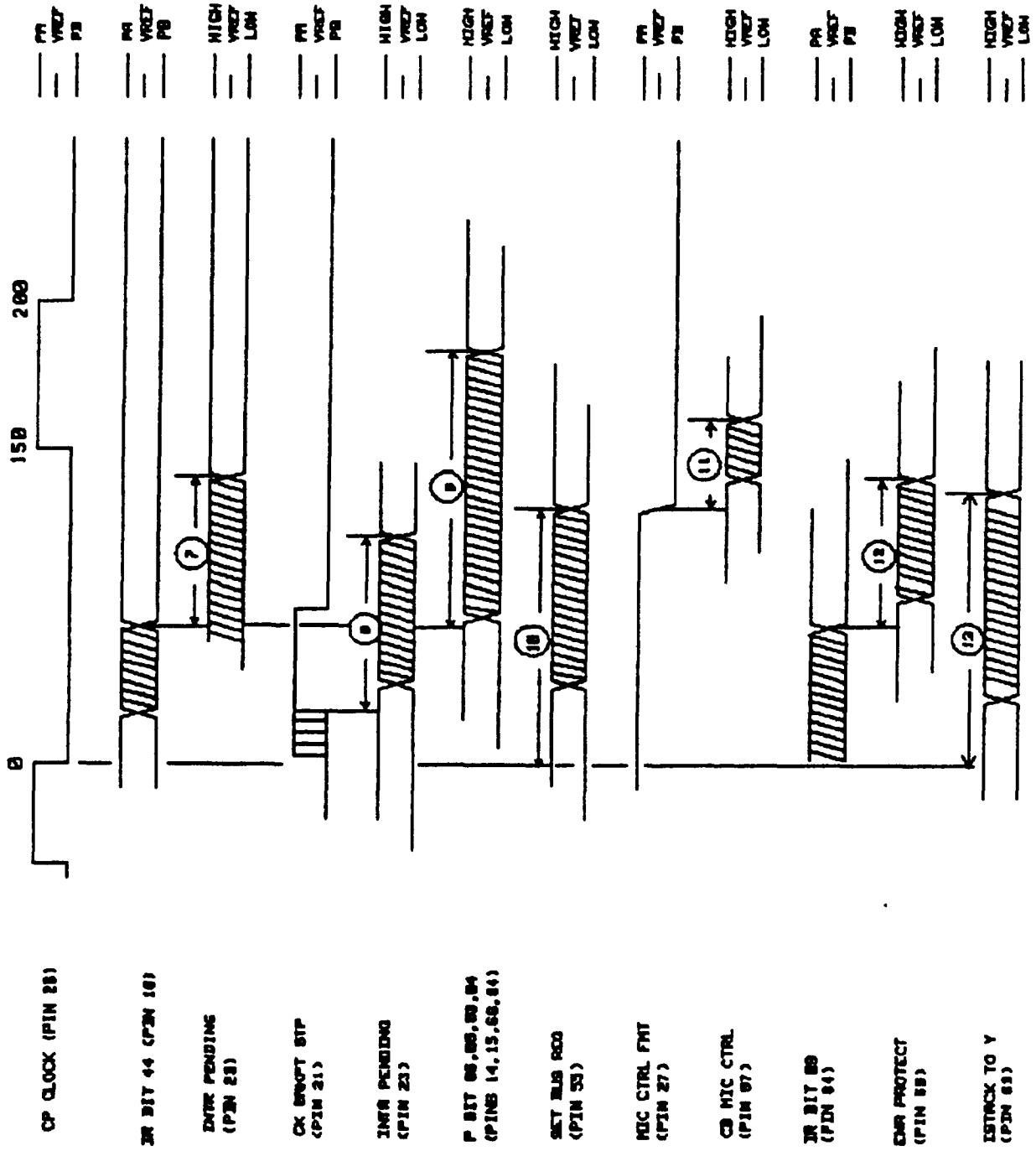


FIGURE 12b. Dynamic waveforms for table XVII.

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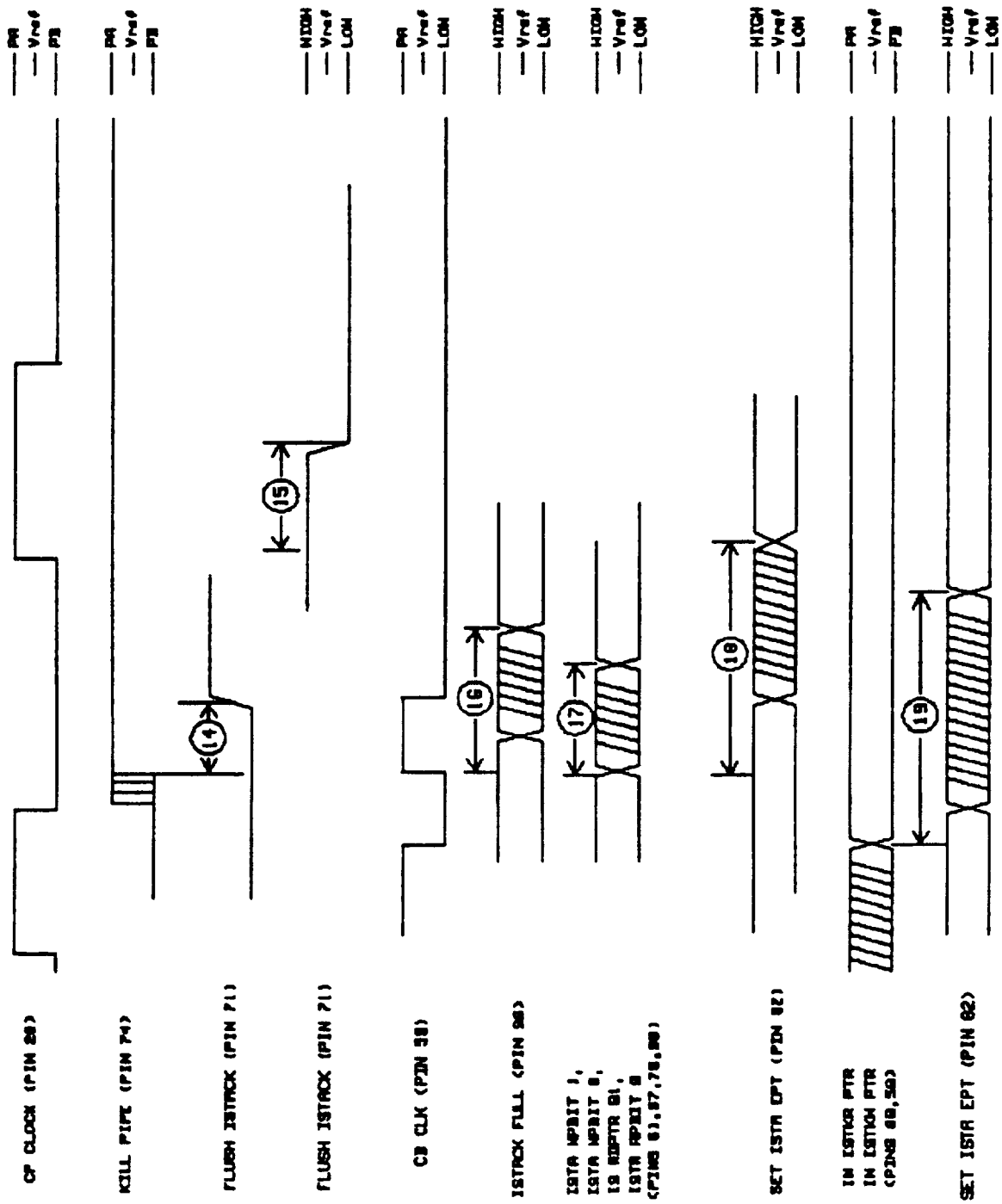


FIGURE 12c. Dynamic waveforms for table XVII.

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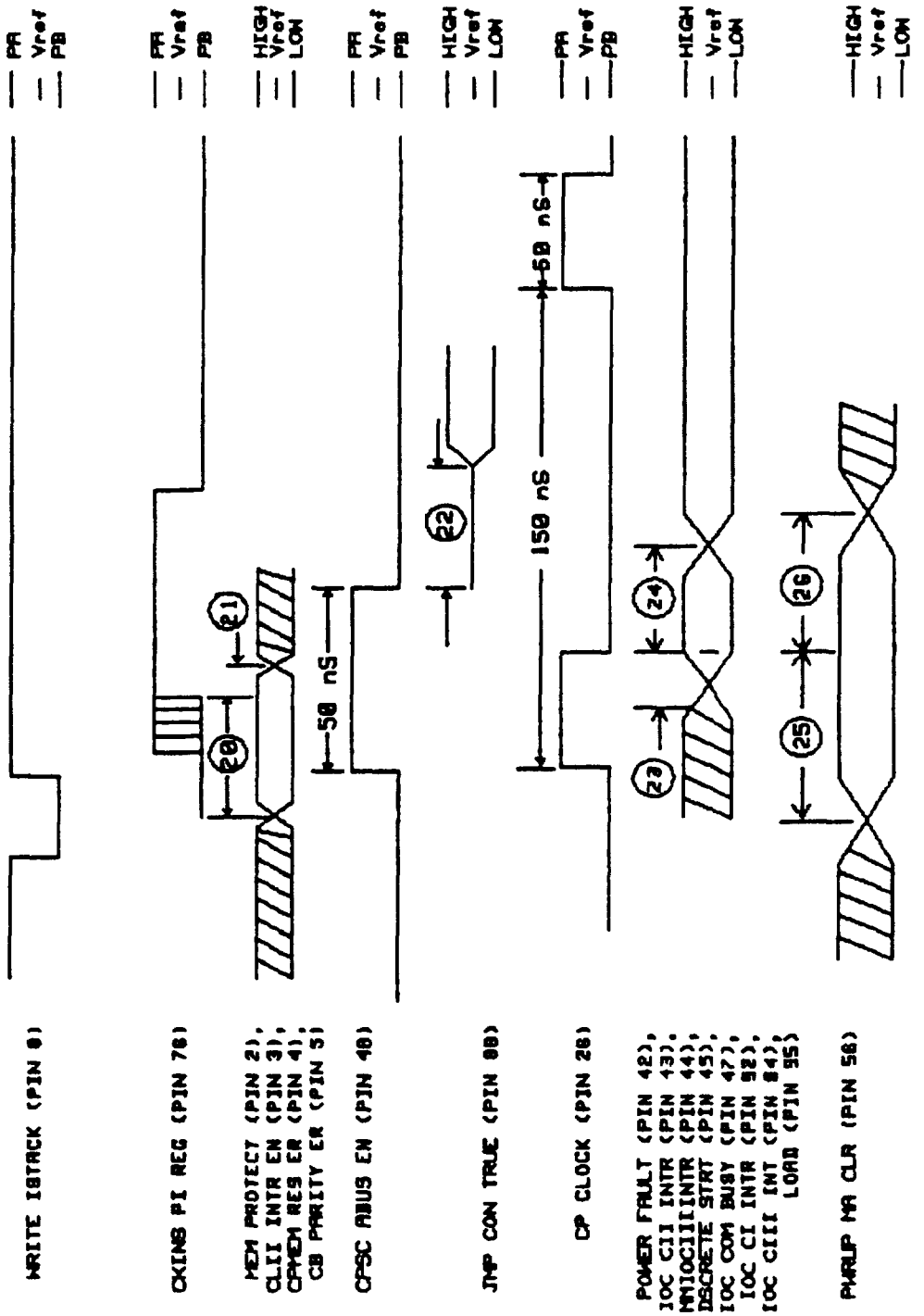


FIGURE 12d. Dynamic waveforms for table XVII.

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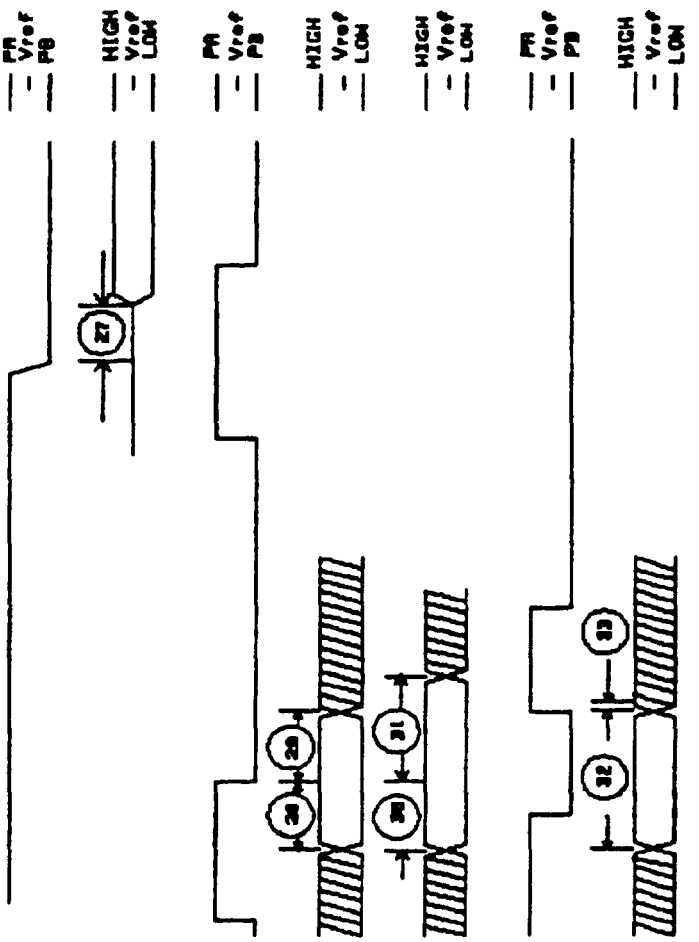


FIGURE 12e. Dynamic waveforms for table XVII.

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TABLE XVIII. Input capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R1	MEM PROTECT	2	4	100	pF	1-4
R2	CLII INTR EN	3	6	125	pF	1-4
R3	CPMEM RES ER	4	2	50	pF	1-4
R4	CB PARITY ER	5	4	125	pF	1-4
R5	DSCRETE MCLR	6	4	125	pF	1-4
R6	WRITE ISTACK	8	2	75	pF	1-4
R7	SEL ISTK SEL	12	2	75	pF	1-4
R8	BREAKPOINT	13	4	75	pF	1-4
R9	MONITOR CLK	16	4	100	pF	1-4
R10	CIII INTR EN	17	2	75	pF	1-4
R11	IR BIT 44	18	8	125	pF	1-4
R12	MPI0MPC VINT	19	2	75	pF	1-4
R13	IR BIT 06	20	2	75	pF	1-4
R14	CK BRKPT STP	21	4	75	pF	1-4
R15	CP CLOCK	26	8	125	pF	1-4
R16	MIC CTRL FMT	27	4	100	pF	1-4
R17	IR BIT 00	28	2	50	pF	1-4
R18	IR BIT 01	29	2	75	pF	1-4
R19	IR BIT 02	30	2	75	pF	1-4
R20	IR BIT 41	32	2	50	pF	1-4
R21	IR BIT 08	33	2	75	pF	1-4

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TABLE XVIII. Input capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R22	IR BIT 10	34	2	75	pF	1-4
R23	IR BIT 13	35	2	75	pF	1-4
R24	IR BIT 15	36	4	75	pF	1-4
R25	MLTMATH PRES	37	4	75	pF	1-4
R26	SGR-EQ TO IR	38	2	75	pF	1-4
R27	BOOT RTC PRS	39	4	50	pF	1-4
R28	CMI PRESENT	41	4	50	pF	1-4
R29	POWER FAULT	42	4	75	pF	1-4
R30	IOC CII INTR	43	4	50	pF	1-4
R31	MMIOCIIIINTR	44	4	75	pF	1-4
R32	DSCRETE STRT	45	4	50	pF	1-4
R33	STOIN ISTK D	46	2	50	pF	1-4
R34	IOC COM BUSY	47	4	75	pF	1-4
R35	CPSC ABUS EN	48	2	100	pF	1-4
R36	JMAP OE	49	2	100	pF	1-4
R37	ECW 13	52	4	100	pF	1-4
R38	ECW 12	53				
R39	ECW 15	54	4	100	pF	1-4
R40	PWRUP MA CLR	56	6	175	pF	1-4
R41	IN ISTKW PTR	58	6	125	pF	1-4
R42	CB CLK	59	6	125	pF	1-4

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TABLE XVIII. Input capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R43	RTC OVFL INT	62	4	125	pF	1-4
R44	IR BIT 45	67	4	125	pF	1-4
R45	CI INTR EN	68	2	100	pF	1-4
R46	DEST BUS	72	2	75	pF	1-4
R47	UIR 5	73	4	75	pF	1-4
R48	KILL PIPE	74	2	75	pF	1-4
R49	CKINS PI REG	76	4	75	pF	1-4
R50	KIL PIPE	77	8	100	pF	1-4
R51	FORC I FETCH	78	4	75	pF	1-4
R52	IN ISTKR PTR	80	8	125	pF	1-4
R53	IR BIT 07	83	2	50	pF	1-4
R54	IR BIT 09	84	2	50	pF	1-4
R55	IR BIT 12	85	2	75	pF	1-4
R56	IR BIT 14	86	2	75	pF	1-4
R57	DSCRT AUTOST	87	4	75	pF	1-4
R58	MAE PRESENT	89	4	75	pF	1-4
R59	IOC PRESENT	91	4	50	pF	1-4
R60	IOC CI INTR	92	4	75	pF	1-4
R61	IOC CIII INT	94				
R62	LOAD	95	4	75	pF	1-4

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NOTES:

1. Input capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	R1-R62
Capacitance meter voltage	15 mV	± 1.5 mV	R1-R62
Capacitance measurement	NA	± 4.8 pF	R1
	NA	± 5.95 pF	R2
	NA	± 2.4 pF	R3
	NA	± 6.05 pF	R4,R5
	NA	± 3.65 pF	R6,R7
	NA	± 3.55 pF	R8
	NA	± 4.8 pF	R9
	NA	± 3.65 pF	R10
	NA	± 5.85 pF	R11
	NA	± 3.65 pF	R12,R13
	NA	± 3.55 pF	R14
	NA	± 5.85 pF	R15
	NA	± 4.8 pF	R16
	NA	± 2.4 pF	R17
	NA	± 3.65 pF	R18,R19
	NA	± 2.4 pF	R20
	NA	± 3.65 pF	R21-R23
	NA	± 3.55 pF	R24,R25
	NA	± 3.65 pF	R26
	NA	± 2.3 pF	R27,R28
	NA	± 3.55 pF	R29
	NA	± 2.3 pF	R30
	NA	± 3.55 pF	R31
	NA	± 2.3 pF	R32
	NA	± 2.4 pF	R33
	NA	± 3.55 pF	R34
	NA	± 4.9 pF	R35,R36
	NA	± 4.8 pF	R37-R39

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Capacitance measurement	NA	± 8.45 pF	R40
	NA	± 5.95 pF	R41,R42
	NA	± 6.05 pF	R43,R44
	NA	± 4.9 pF	R45
	NA	± 3.65 pf	R46
	NA	± 3.55 pF	R47
	NA	± 3.65 pF	R48
	NA	± 3.55 pF	R49
	NA	± 4.6 pF	R50
	NA	± 3.55 pF	R51
	NA	± 5.85 pF	R52
	NA	± 2.4 pF	R53,R54
	NA	± 3.65 pF	R55
	NA	± 4.9 pF	R56
	NA	± 3.55 pF	R57,R58
	NA	± 2.3 pF	R59
	NA	± 3.55 pF	R60-R62

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVIII.

3. The test circuit shall be as indicated on figure 13.

4. Abbreviations used are defined in 6.1.

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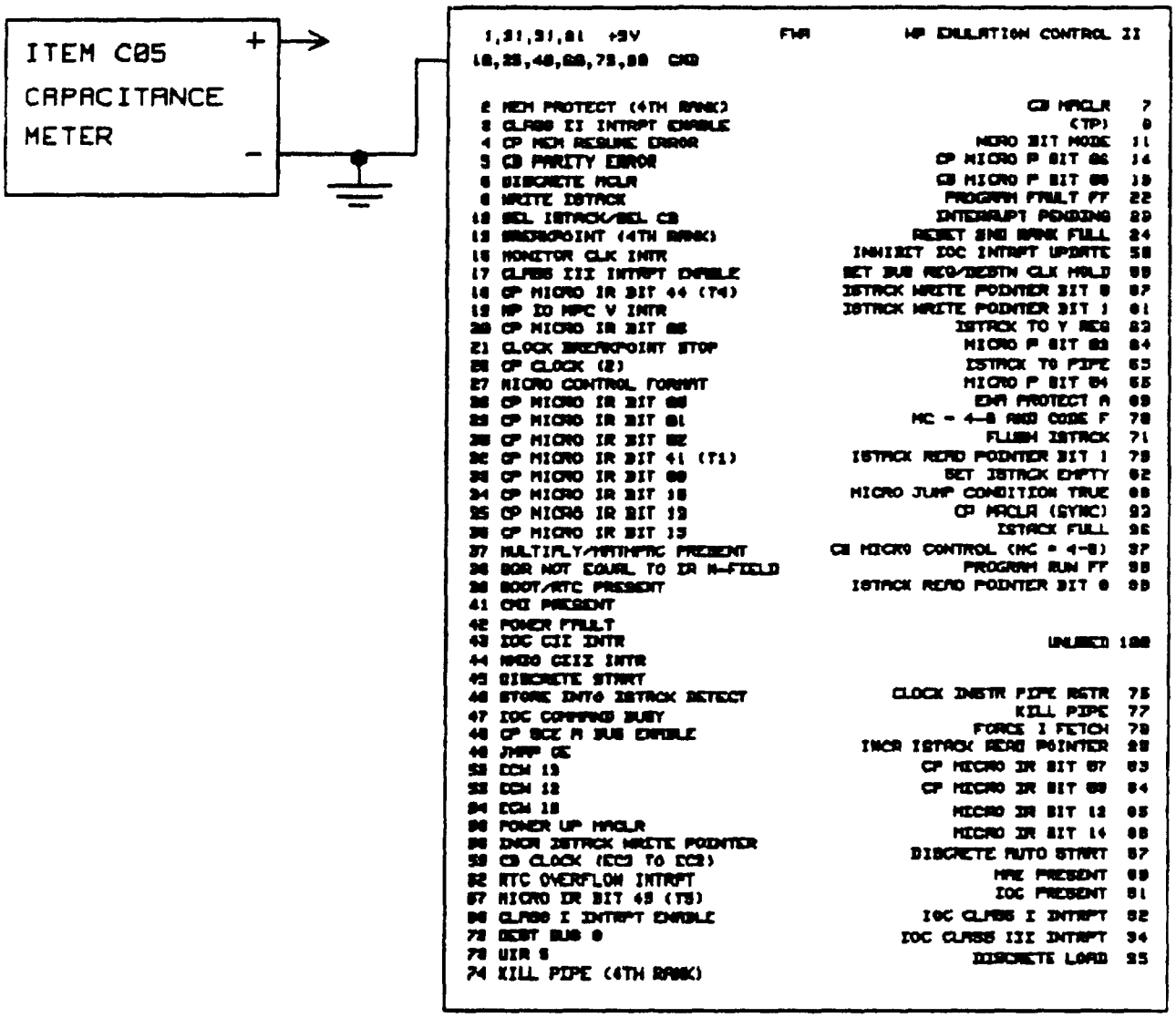


FIGURE 13. Input capacitance test circuit for table XVIII.

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TABLE XIX. Input clamp diode latch up test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
S1	IN ISTKW PTR	58	-18.0	-1.2	2.4	V	1-3
S2	CB CLK	59					
S3	IN ISTKR PTR	80	-18.0	-1.2	2.4	V	1-3

NOTES:

1. Input clamp diode latch up test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	S1-S3
Power supply current limit	1.93 A	NA	S1-S3
Forcing current, (Iin)	-18 mA	± 1.8 mA	S1-S3
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	S1-S3 S1-S3
Input voltage measurement	NA	± 180 mV	S1-S3

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XIX and the voltage then measured.

(d) The forcing pulse shall have a duration of 500 $\pm$ 20 nS.

3. Abbreviations used are defined in 6.1.

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TABLE XX. Power supply transient amplitude test.

Test cond	Parameter	Pin number	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs
T1	PSTA	1	0	400	mV	1-4	14,15
T2	PSTA	51	0	400	mV	1-4	14,15

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	T1,T2
Power supply current limit	1.93 A	NA	T1,T2
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 T1,T2 T1,T2
Power supply transient amplitude measurement	NA	± 20 mV	T1,T2

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XX.

(c) Vcc pins 1 and 31 shall be shorted together, pins 51 and 81 shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XX.

(d) Using the oscilloscope, D06, inspect for the most negative spike as shown on figure 14 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. The test circuit shall be as indicated on figure 15.

4. Abbreviations used are defined in 6.1.

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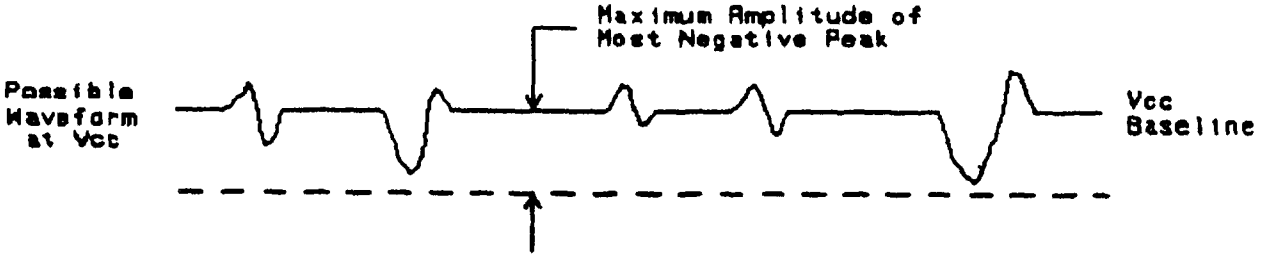
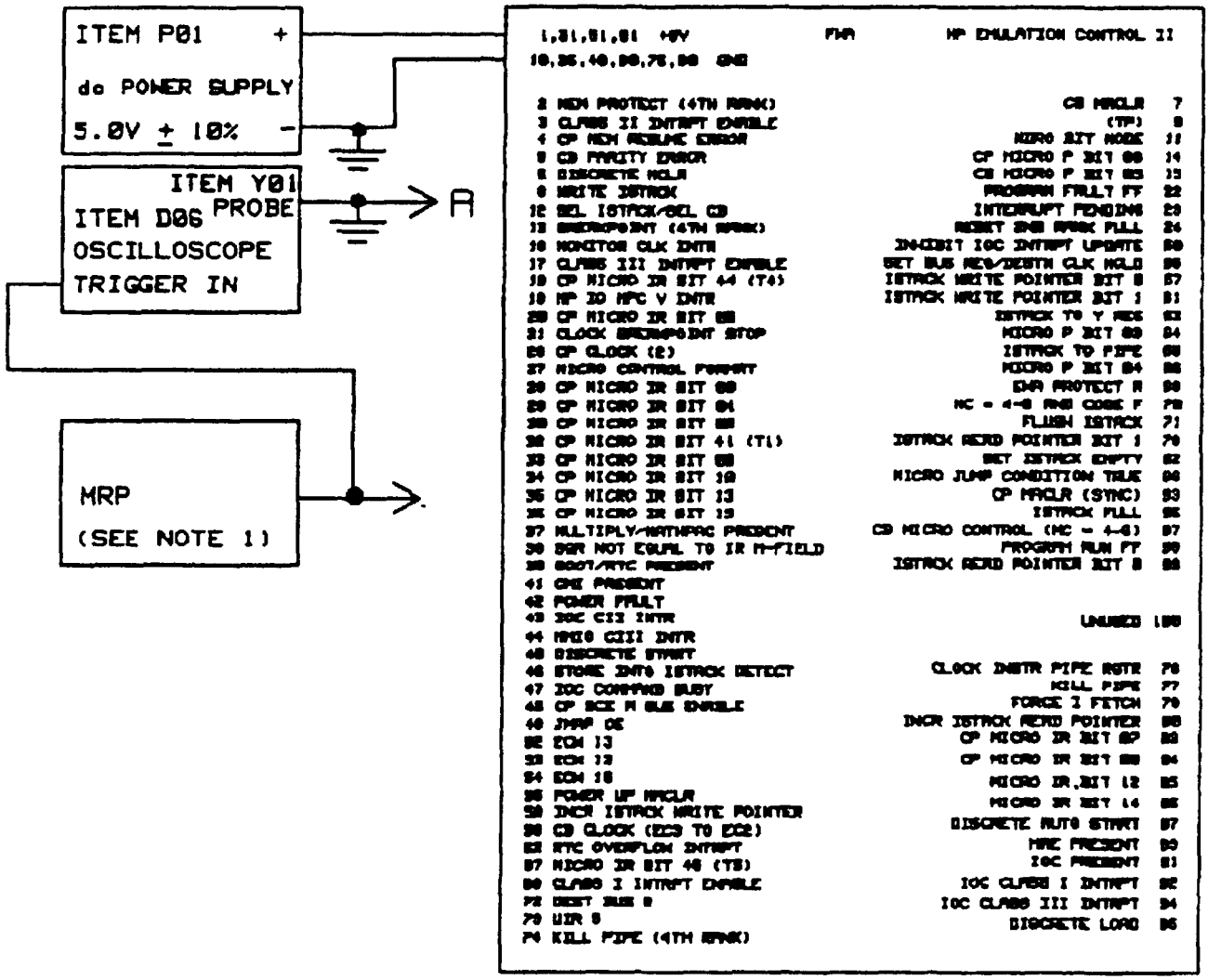


FIGURE 14. Power supply transient amplitude waveform for table XX.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 15. Power supply transient amplitude test circuit for table XX.

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5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording.

c Reason/Rationale for Recommendation·

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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